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KISS

Keep bIfurcation Single stenting Simple

Sponsor: CERIC (Center for European Research Initiatives in Cardiovascular Medicine)

Investigational device(s): Medtronic Resolute Onyx stent

Clinical Investigation Report	
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NOTICE

The Clinical investigation was conducted in accordance with the ISO14155 standard requirements and applicable regulatory requirements.

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1 Summary

Title of the clinical investigation - summary information	
Brief Study Title	KISS
Full study title	Keep bIfurcation Single stenting Simple
Dates of investigation	Clinical investigation initiation: 09/10/2020 (first patient in) Clinical investigation completion: 18/12/2023 (last patient last visit)

Locations

Country	Site name and address	Principal Investigator
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Country	Site name and address	Principal Investigator
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Purpose of the clinical investigation
The objective was to evaluate the non-inferiority of no side branch (SB) intervention versus side branch ballooning, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion.

Description of the investigational device, clinical investigation, and methods used	
Description of participants	KISS study was intended to include patients with coronary heart disease particularly patients with **de novo** coronary bifurcation lesion with >70% lesion in the main vessel requiring revascularization. <u>Subject were included if all of the following criteria were met:</u> 1. Coronary artery disease requiring revascularization that is amenable to PCI; 2. Any **de novo** Medina type bifurcation lesion with >70% lesion in the main vessel by visual assessment except Medina 0,0,1 if the planned strategy does not include a main branch stenting; 3. Side branch diameter compatible with a PCI with stent implantation (≥ 2.25mm); 4. Side branch requiring a protection wire according to operator evaluation; 5. Anatomy compatible with a proximal optimization technique; 6. Post main branch treatment (stenting then POT): Absence of side branch occlusion or low flow associated with ongoing ischemia (either chest pain or ST changes); 7. Male or female patient >18 years old. <u>Subject were not included if any one of the following conditions existed:</u> 1. Acute myocardial infarction with ongoing ST-elevation; 2. Cardiogenic shock; 3. Requirement for ongoing hemodialysis; 4. Life-expectancy limited to <12 months due to co-morbid condition; 5. Lesion involved the left main coronary artery; 6. Lesion required a double stenting; 7. Plan to treat >1 other coronary vessel at the time of the index procedure; 8. Two lesions to be treated in the same session: unsuccessful PCI on the single non-bifurcation lesion in a non-target vessel before the treatment of the target

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	lesion; 9. Chronic total occlusion of any target vessel; 10. Left ventricular ejection fraction <20%; 11. Side branch TIMI Flow <3; 12. Pre-dilatation of the Side Branch during the procedure prior to randomization; 13. Known allergy to Aspirin; 14. Known allergy to Clopidorel and Prasugrel and Ticagrelor; 15. Known allergy to stent drug elutant; 16. Known allergy to any other component of Onyx Resolute stent; 17. Ongoing participation in another investigational device or drug study; 18. Inability to provide informed consent; 19. Patients under judicial protection, tutorship or curatorship; 20. Pregnant and breast-feeding women or intention to become pregnant prior to completion of all follow-up procedures.
Description of the device	The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) consists of a balloon-expandable, intracoronary, drug-eluting stent (DES) premounted on a Rapid Exchange (RX) or an Over-the-Wire (OTW) stent delivery system. The Resolute Onyx™ stent is manufactured from a composite material of cobalt alloy and platinum-iridium alloy and is formed from a single wire bent into a continuous sinusoid pattern and then laser fused back onto itself. The stents are available in multiple lengths and diameters.
Description of procedures to use the device	The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is indicated for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to de novo lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameters of 2.25 mm to 5.0 mm.
Study Design	KISS study is an investigator-initiated, multi-centre, prospective, randomized (1:1), parallel, two-arm, non-inferiority trial which aimed to compare two bifurcation PCI

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	procedures for SB protection. KISS study was completed in 6 countries where Medtronic Resolute Onyx stent was CE approved and commercially used.
Objectives and endpoints	The primary objective of KISS study was to evaluate the non-inferiority of no side branch (SB) intervention: **test arm**, versus side branch ballooning: **control arm**, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion. **<u>Primary Endpoint</u>** Rate of peri-procedural MI (PPMI) at discharge or 48h post-procedure whichever comes first as defined by the ARC consortium (Academic Research Consortium, so called "ARC-2" definitions) and based on post-PCI troponin rise as follows: **Periprocedural MI PCI** (within 48 Hours)**:** Absolute rise in cardiac troponin (hs-cTn: high-sensitivity cTn) from baseline ≥35 times upper reference limit +/and one (or more) of the following criteria: - New significant Q waves or equivalent; - Flow-limiting angiographic complications; - New "substantial" loss of myocardium on imaging. **Significant periprocedural myocardial injury:** Absolute rise in cardiac troponin (hs-cTn) from baseline ≥70 times upper reference limit **<u>Secondary Endpoints</u>** 1) <u>Procedural Endpoints</u> • Technical success: deployment of main branch stent with <20% residual stenosis and a patent side branch at end of procedure upon Corelab analysis; • Total procedure time; • Acute gain in main and side branches as measured with a dedicated QCA software; • Fluoroscopy time; • Total radiation dose (air kerma);

	• Total contrast dose;
	• Need to cross over *(any intervention on SB in the test arm or failure to intervene on the SB in the control arm).*
	
	2) <u>Clinical endpoints</u>
	The following endpoints were evaluated at 1-month and at 12-months.
	• Target Lesion Failure (TLF) defined as the composite of:
	  ○ Cardiac Death;
	  ○ Target vessel Myocardial Infarction;
	  ○ Clinically driven Target lesion revascularization (ci-TLR).
	• Main Branch (MB) & Side Branch (SB) TLR;
	• Definite or Probable stent thrombosis (ARC-2 criteria);
	• Angina status according to the CCS classification (Grade 1 to 4).

Sample size	596 patients Assuming a rate of 15% Myocardial Infarction (PPMI) as defined by ARC-2 criteria in the control group, taking a 6% drop out rate into account and using an absolute non-inferiority limit of 7.5%, it was calculated that 596 subjects randomized equally between the 2 arms would have 80% power to show non-inferiority at a one-sided significance level of 0.05.
Randomisation and blinding	Patients were randomized 1:1 using KISS e-CRF during the PCI after main branch stenting to either strategies: • No intervention on the Side Branch; • Side Branch ballooning or Kissing Balloon Technique - KBT (left at operator discretion) using a non-compliant 1:1 balloon Neither the patients nor the physicians were blinded.
Follow up duration	12 months
Concomitant treatments	*NA*
Statistical analysis methods	SAS Software 9.4 was used to conduct data processing and statistical analyses. The difference between the two arms regarding the primary

	endpoint was analyzed using a Farrington-Manning Score Test. The differences between the two arms regarding TLF, main branch TLR, side branch TLR and definite or probable stent thrombosis were done by presenting the risk differences with a two-sided 95% CI and a Z-test. The difference between the two arms regarding the proportion improvers in angina status from baseline to 1-month and 12-months, the technical success and need to cross over were analysed using a chi square test, or Fisher's exact test if there were cells with expected counts lower than 5 patients. The difference between the two arms regarding the total procedure time, acute gain in main and side branches, fluoroscopy time, total radiation dose and total contrast dose were analysed using a 2-sample t-test, or a Wilcoxon rank-sum test if the data shows serious deviations from a normal distribution.
Substantial modifications	Prolongation of recruitment period (12 months)

Results of the investigation	
Participant flow	A total of 617 patients were randomized in the study. Twenty additional patients, compared to the initially calculated sample size, were included due to unreported data affecting the assessment of primary endpoint for approximately 20 patients. The follow-up rate was 99.5% at 30 days and 98.7% at 12 months.
Baseline demographic and clinical characteristics (Intention To Treat Population)	The average age of the patients was 67.4 years in the no-SB intervention group and 68.0 years in the SB intervention group. Among the patients, 75.9% were male in the no-SB intervention group and 78.6% in the SB intervention group. At baseline, 7.6% of patients in the no-SB intervention group and 4.5% in the SB intervention group had a history of cerebrovascular disease. Additionally, 26.4% of patients in the no-SB intervention group and 20.4% in the SB intervention group had a family history of coronary disease. Previous coronary intervention (PCI) was reported in 28.0% of patients in the no-SB intervention group and 37.0% in the SB

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	intervention group. Previous myocardial infarction (MI) was present in 13.5% of patients in the no-SB intervention group and 28.5% in the SB intervention group, while previous coronary artery bypass grafting (CABG) was noted in 0.7% of patients in the no-SB intervention group and 0.3% in the SB intervention group. Regarding lifestyle and comorbidities, 22.1% of patients in the no-SB intervention group and 20.1% in the SB intervention group were smokers. Diabetes was reported in 27.0% of patients in the no-SB intervention group and 27.6% in the SB intervention group, while hypertension was present in 63.0% of patients in the no-SB intervention group and 61.0% in the SB intervention group. Hypercholesterolemia was observed in 52.8% of patients in the no-SB intervention group and 57.5% in the SB intervention group. There were no significant differences in baseline demographic or clinical characteristics between the two groups, except for previous PCI and previous MI, which were significantly more frequent in the SB intervention group.
Outcome of the intervention	Primary endpoint defined as rate of PPMI at discharged or 48 hours, was 4.1% in no SB intervention group and 5.9% in SB intervention group ($p < 0.001$ for non inferiority and p = 0.384 for superiority). Technical success rate of the bifurcation lesion treatment was 80.2% in no SB intervention group and 82.1% in SB intervention group with no significant difference ($p > 0.544$). Procedure Time was significantly lower in no SB intervention group 34 minutes compared to SB intervention group 45 minutes ($p < 0.001$). No SB intervention was also associated with significant reduction in contrast media volume (130mL *vs* 150mL, $p < 0.001$), X-ray exposure (Air Kerma 453mGy *vs* 629 mGy, $p < 0.001$) and a less requirement for a double stenting (0.3% *vs* 4.5%, $p < 0.01$).
Safety outcomes	TLF rate was 3.6% at 30 days and 5.0% at 12months in no SB intervention group and 4.5% at 30 days and 6.4% at 12months in SB intervention group, with no significant difference. At 12 months, there was no significant difference between no SB and SB intervention group in the rates of death (2.0% *vs*

	2.3%), Clinical driven TLR (1.3% *vs* 1.9%), Definite or Probable ST (0.7% in both arm), and vessel MI (4.0% *vs* 4.8%).
Limitations	This open-label design may have introduced operator bias, though core lab review mitigated this risk. The inclusion criteria (bifurcation lesions with preserved SB flow) limit generalizability to more complex anatomies. The ARC-2–based periprocedural MI definition may be influenced by procedural variability and differences across centres and assays. Short follow-up and lower-than-expected MI rates reduced statistical power. Finally, PCI was not guided by intracoronary imaging, which later evidence and guidelines have shown to improve outcomes.
Conclusion	The management of the SB during bifurcation PCI remains challenging. A conservative strategy without systematic SB intervention in case of normal SB flow is associated with very rare procedural complications and is non inferior to a systematic SB intervention regarding periprocedural MI and injuries. These findings support a simplified "Keep Bifurcation Stenting Simple" approach in non-complex and non-LM bifurcation PCI. Further studies are warranted to support these data and evaluate the impact of this conservative strategy on long-term clinical events.
Potential for future studies	Further studies are warranted to evaluate the impact of this conservative strategy on long-term clinical events, the predictive factors of side branch flow impairment and the extrapolation of these data in all bifurcation lesions including left main.
Deviations to clinical investigation plan	The most significant deviation from the clinical investigation plan observed during the trial was the failure to assess cardiac enzymes before or after the procedure. Other protocol deviations, although rare, included side branch (SB) occlusion or slow flow following main branch (MB) stenting, as well as the omission of proximal optimization technique (POT) or kissing balloon inflation.

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2 Investigators and administrative structure of clinical investigation

2.1 Coordinating/Principal investigator

The KISS study is an investigator-initiated trial conducted independently by the investigators.

The coordinating investigator of the KISS study was Dr Bernard Chevalier
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2.2 Sponsor

CERIC located in (Villars-sur-Glâne, Switzerland) sponsored the clinical investigation. **Table 1** describes the key Sponsor personnel that monitored and oversaw the clinical investigation.

Table 1: Key Sponsor personnel

Role	Contact Information
General Manager	Prof Stéphane COOK CERIC 1752 Route des Dailles 2 Villars-sur-Glâne Switzerland scook@ceric-europe.org

2.3 Grant Giver

The KISS study was entirely financed by Medtronic.

2.4 Contract Research Organization

The KISS study was managed by the CERC (Cardiovascular European Research Center) based in Massy, France.

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CERC was responsible for:

- Sites selection
- Study documents writing
- Regulatory Submissions
- Hospital Contracts Management
- Monitoring
- Safety Reporting
- Core Laboratory
- CEC activities
- Data Management & Statistical activities
- Clinical Investigation Report writing

2.5 Clinical Events Committee

Table 2 lists the Clinical Events Committee (CEC) members involved in the clinical investigation.

Table 2: Chairperson and members

Role	Contact Information
CEC Chairperson	Prof Emanuele Barbato (Sapienza, Universita Di Roma) Rome, Italy
CEC Member	Dr Fina Mauri Barcelona, Spain
CEC Member	Dr Yves Louvard Guérande, France

3 Introduction

Coronary stenting takes a central place to the current therapeutic management of coronary artery disease (CAD) in case of percutaneous coronary intervention (PCI) for coronary revascularization. A coronary bifurcation is involved in 15 to 20% of the patients(1) and induces some technical challenges including artery diameter incongruency, risk of stent malapposition leading to higher risk of stent thrombosis(2) and particularly risk of side branch (SB) impairment mainly due to bifurcation carina shift or dissection. In order to

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maintain the SB patency, many techniques have been proposed(3) but provisional stenting strategy has been established as a standard of care in most of the bifurcation lesions (4).

In order to comply with the fractal law of the coronary tree(5), it is recommended to select a stent sized to distal reference diameter and to systematically achieve a proximal optimization technique (POT) allowing a correct apposition of the proximal part of the stent and a scaffolding of the SB ostium (4,6–8). Once these steps are completed, the clinical benefit of an additional intervention on SB ostium is a matter of debate. A SB intervention is expected to decrease the risk of occlusion and, subsequently, of peri-procedural myocardial injury and related worse outcomes (9,10). A SB intervention could be also expected to improve angina relief by reducing the risk of SB lesion evolution and requiring reintervention. On the contrary, the absence of SB intervention could reduce procedure time, contrast media volume and X-ray exposure as the risk of SB dissection and second stent implantation leading to potential higher risk of stent thrombosis (6).

Several registries and small randomized clinical trials addressed this issue but most of them did not take into account the modern strategy of provisional stenting including absence of systematic SB predilatation, stenting of the main branch (MB) sized to distal reference and systematic POT (11–14). In these studies, SB intervention was mainly a traditional kissing balloon inflation (KBI) (15) without considering the recent strategy of POT/side branch ballooning/POT sequence proposed as an alternative due to its quicker execution and potential benefit on outcomes (16,17).

As a consequence, the impact on procedural success and long-term outcomes of systematic SB intervention based on the contemporary optimized coronary bifurcation revascularization is unknown. The main objective of the present randomized multicenter KISS trial was to determine whether a conservative strategy only based on revascularization of the main branch is non-inferior to a systematic SB intervention.

4 Investigational device

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) is a device/drug combination product comprised of the following device components: the Resolute Onyx™ coronary stent and delivery system and a drug component (a formulation of zotarolimus in a polymer coating).

Medtronic Resolute Onyx stent obtained its CE (Conformité Européenne) mark in 2014 and a U.S Food and Drug Administration (FDA) Approval in 2017 (18). Complementary information to the following sections can be found in Medtronic Resolute Onyx Stent Instructions for Use (IFU).(19)

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4.1 Device Component Description

The Medtronic Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System (Resolute Onyx™ system) consists of a balloon-expandable, intracoronary, drug-eluting stent (DES) premounted on a Rapid Exchange (RX) or an Over-the-Wire (OTW) stent delivery system.

The Resolute Onyx™ stent is manufactured from a composite material of cobalt alloy and platinum-iridium alloy and is formed from a single wire bent into a continuous sinusoid pattern and then laser fused back onto itself. The stents are available in multiple lengths and diameters. The delivery system has two radiopaque markers to aid in the placement of the stent during fluoroscopy and is compatible with 0.014-inch (0.36-mm) guidewires and 1.42-mm (5-Fr/0.056-in) minimum inner diameter guide catheters. The Resolute Onyx™ RX delivery system and the Resolute Onyx™ OTW delivery system have an effective length of 140 cm. (19)

4.2 Drug Component Description

The drug coating of Resolute Onyx™ System consists of the drug Zotarolimus (the active ingredient) and BioLinx® polymer system (the inactive ingredient). (19)

Zotarolimus
The active pharmaceutical ingredient utilized in the Resolute Onyx™ System is zotarolimus. It is a tetrazole-containing macrocyclic immunosuppressant. Zotarolimus has extremely low water solubility and is a lipophilic compound that is freely soluble in Propylene glycol, Acetone, Toluene, Acetonitrile, Ethanol, Benzyl alcohol and Dimethyl sulfoxide (DMSO). The molecular formula of zotarolimus is C52H79N5O12 and its molecular weight is 966.2. Zotarolimus does not have any ionizable group(s) in the physiological pH range; therefore, its solubility is expected to be unaltered in this range.

Polymer System Description
The Resolute Onyx™ stent is comprised of a bare metal stent (BMS) with a Parylene C primer coat and a coating that consists of a blend of the drug zotarolimus and the BioLinx® polymer system. BioLinx® is a blend of the Medtronic proprietary components C10 and C19, and PVP (polyvinyl pyrrolidone).

4.3 Indications

The Resolute Onyx™ Zotarolimus-Eluting Coronary Stent System is indicated for improving coronary luminal diameters in patients, including those with diabetes mellitus, with symptomatic ischemic heart disease due to de novo lesions of length ≤ 35 mm in native coronary arteries with reference vessel diameters of 2.25 mm to 5.0 mm.(19)

4.4 Contraindications

The Resolute Onyx™ System is contraindicated for use in: (19)

- Patients with known hypersensitivity or allergies to aspirin, heparin, bivalirudin, clopidogrel, prasugrel, ticagrelor, ticlopidine, drugs such as zotarolimus, tacrolimus, sirolimus, everolimus, or similar drugs or any other analogue or derivative.
- Patients with a known hypersensitivity to the cobalt-based alloy (cobalt, nickel, chromium, and molybdenum) or platinum-iridium alloy.
- Patients with a known hypersensitivity to the BioLinx® polymer or its individual components

Coronary artery stenting is contraindicated for: (19)

- Patients for whom anti-platelet and/or anticoagulation therapy is contraindicated.
- Patients with lesions that prevent the complete inflation of an angioplasty balloon or the proper placement of the stent or stent delivery system.

5 Clinical Investigation Plan

5.1 Clinical investigation objectives

The primary objective of KISS study was to evaluate the non-inferiority of no side branch (SB) intervention: **test arm**, versus side branch ballooning: **control arm**, in the setting of single stenting with systematic proximal optimization technique (POT), in the treatment of bifurcation lesion.

a. Primary Endpoint(s)

Rate of peri-procedural MI **at discharge or 48h post-procedure** whichever comes first as defined by the ARC consortium (Academic Research Consortium, so called “ARC-2” definitions) (20) and based on post-PCI troponin rise as follows:

Periprocedural MI PCI (within 48 Hours)**:**

Absolute rise in cardiac troponin (hs-cTn: high-sensitivity cTn) from baseline ≥35 times upper reference limit

+/and one (or more) of the following criteria:

- New significant* Q waves or equivalent

- Flow-limiting angiographic complications
- New “substantial” loss of myocardium on imaging

Significant periprocedural myocardial injury:
Absolute rise in cardiac troponin (hs-cTn) from baseline ≥70 times upper reference limit

*These definitions were applicable in patients with normal (or elevated and stable or falling) baseline biomarkers. *Q-wave criteria requires the development of new Q waves ≥40 ms in duration and ≥1 mm deep in voltage in ≥2 contiguous leads.*

b. Secondary Endpoints

Procedural Endpoints:

- Technical success: deployment of main branch stent with <20% residual stenosis and a patent side branch at end of procedure upon Corelab analysis (additional information available in the Corelab Plan, Appendix 2)
- Total procedure time
- Acute gain in main and side branches as measured with a dedicated QCA software
- Fluoroscopy time
- Total radiation dose (air kerma)
- Total contrast dose
- Need to cross over *(any intervention on SB in the test arm or failure to intervene on the SB in the control arm)*

Clinical endpoints:

The following endpoints were evaluated at 1-month and at 12-months:

- Target Lesion Failure (TLF) defined as the composite of:
  - Cardiac Death,
  - Target vessel Myocardial Infarction,
  - Clinically driven Target lesion revascularization (ci-TLR).

  Target lesion is the segment where coronary intervention was performed, and the length of the target lesion is inclusive of the arterial segment treated with the stent and the 5mm proximal and 5mm distal to the treated section.

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- Main Branch (MB) & Side Branch (SB) TLR
- Definite or Probable stent thrombosis (ARC-2 criteria) (20)
- Angina status according to the CCS classification (Grade 1 to 4) (21)

5.2 Clinical investigation design

KISS study was, multi-centre, prospective, randomized (1:1), parallel, two-arm, non-inferiority trial aiming to compare two bifurcation PCI procedures for SB protection, performed in 6 countries where Medtronic Resolute Onyx stent is CE approved and commercially used. Kiss study was designed to reflect the standard of care routinely practiced in the participating hospitals.

5.3 Clinical investigation population

KISS study intended to include patients with coronary heart disease particularly patients with **de novo** coronary bifurcation lesion with >70% lesion in the main vessel requiring revascularization.

Patients should had no more than 1 additional coronary lesion (excluding a bifurcation lesion) not on the target vessel requiring treatment, such that the main focus of the study remain on the bifurcation disease. The lesion which had to be treated with PCI should have been treated preferably before, the index lesion. In the case of stable patients this allowed randomisation to occur after treatment of non-index lesion.

If other lesions had to be treated, drug-eluting stents (Medtronic Resolute Onyx) were to be used to reduce risk of subsequent revascularisation unrelated to the index bifurcation disease.

5.3.1 Inclusion criteria

Subject were included if all of the following criteria were met:

1. Coronary artery disease requiring revascularization that is amenable to PCI
2. Any **de novo** Medina type bifurcation lesion with >70% lesion in the main vessel by visual assessment except Medina 0,0,1 if the planned strategy does not include a main branch stenting
3. Side branch diameter compatible with a PCI with stent implantation (≥ 2.25mm)
4. Side branch requiring a protection wire according to operator evaluation
5. Anatomy compatible with a proximal optimization technique

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6. Post main branch treatment (stenting then POT): Absence of side branch occlusion or low flow associated with ongoing ischemia (either chest pain or ST changes)
7. Male or female patient >18 years old

5.3.2 Exclusion criteria

Subject were not included if any one of the following conditions existed:

1. Acute myocardial infarction with ongoing ST-elevation
2. Cardiogenic shock
3. Requirement for ongoing hemodialysis
4. Life-expectancy limited to <12 months due to co-morbid condition
5. Lesion involves the left main coronary artery
6. Lesion requiring a double stenting
7. Plan to treat >1 other coronary vessel at the time of the index procedure
8. Two lesions to be treated in the same session: unsuccessful PCI on the single non-bifurcation lesion in a non-target vessel before the treatment of the target lesion
9. Chronic total occlusion of any target vessel
10. Left ventricular ejection fraction <20%
11. Side branch TIMI Flow <3
12. Pre-dilatation of the Side Branch during the procedure prior to randomization
13. Known allergy to Aspirin
14. Known allergy to Clopidogrel and Prasugrel and Ticagrelor
15. Known allergy to stent drug elutant
16. Known allergy to any other component of Onyx Resolute stent
17. Ongoing participation in another investigational device or drug study
18. Inability to provide informed consent
19. Patients under judicial protection, tutorship or curatorship
20. Pregnant and breast-feeding women or intention to become pregnant prior to completion of all follow-up procedures

5.4 Treatment/Procedure

5.4.1 Enrollment Process

Consecutive patients scheduled for bifurcation PCI were screened for eligibility. Patients who met all clinical inclusion criteria and did not meet any clinical exclusion

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criteria were informed about the study's purpose and procedures, as detailed in the informed consent form (Appendix 3), and were invited to participate.

Written informed consent was obtained prior to enrollment. During the procedure, once procedural eligibility was confirmed (including stenting of the treatable lesion, absence of SB predilatation and absence of SB occlusion or SB slow flow), patients were randomized in a 1:1 ratio through the KISS e-CRF system, immediately after Main Branch stenting, to one of two strategies (Figure 1):

(i) no intervention on the Side Branch (Option 1), or
(ii) Side Branch treatment at the operator's discretion, consisting of ballooning (Option 3) or the Kissing Balloon Technique (Option 2) using a non-compliant 1:1 balloon.

Figure 1. Post-Main Branch stenting strategies

5.4.2 Study Handling Procedures

The total follow-up duration for each participant was 12 months. All examinations, procedures, and assessments were performed according to the standard practice of

the hospital where the participant was treated, in line with current therapeutic recommendations. Data were collected at predefined assessment visits: Baseline (pre-procedure), Procedure, Day 1 and hospitalization, Discharge, 1 month, and 12 months.

The following paragraphs outline the study schedule of assessments, as defined in the protocol, specifying the procedures performed at each visit and the data collected during follow-up.

- Baseline

Prior to PCI, and in accordance with international guidelines (ACCF/AHA/SCAI (22), ESC/EACTS (23)) as well as the stent manufacturer's instructions (19), all participants were to receive dual antiplatelet therapy (DAPT).

- If not already established on Aspirin 75 mg daily for at least five days before PCI, a loading dose of 250 mg was to be administered on the morning of the procedure.
- Dual therapy with Clopidogrel, Prasugrel, or Ticagrelor was acceptable. In the absence of prior treatment for at least five days, loading doses were recommended: Clopidogrel 600 mg, Prasugrel 60 mg, or Ticagrelor 180 mg, administered at least two hours prior to PCI (22,23).

Before revascularization, a full clinical assessment was conducted, including review of medical history and angina status (CCS classification (21)). Routine blood tests were performed per local practice, including measurement of troponin levels (ideally within 24 hours prior to PCI), together with a 12-lead ECG.

- Procedure

PCI was performed according to routine practice, with choice of access site, haemostatic technique, and closure devices left to the operator's discretion. Activated clotting time (ACT) was recommended to be maintained >250 seconds during the procedure. If feasible, PCI of a single non-bifurcation lesion in a non-target vessel could be performed during the same session, provided it was completed before treatment of the target lesion.

For bifurcation PCI, the recommended procedural strategy included:

- Wiring both limbs of the bifurcation after at least two baseline angiographic views, ensuring minimal overlap and clear visualization of the side branch (SB) ostium.

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- Preparation of the main branch (MB) lesion at operator discretion, followed by stent implantation (Resolute Onyx), sized 1:1 with the distal MB reference diameter. Stent length was to allow for a proximal optimization technique (POT) with a balloon-to-artery ratio of 1:1.
- **SB predilatation was prohibited and would result in participant exclusion.**
- After POT, and in the absence of SB occlusion or slow flow with ongoing ischemia (chest pain or ST-segment changes), participants were randomized to one of the two treatment arms (see Section 5.4.1). A wire was to remain jailed in the non-stented vessel until randomization.
- In the KBT arm, a final POT was recommended, and mandatory if ballooning of the SB had been performed ("POT-Side-POT" technique).

Adjunctive intravascular imaging (IVUS or OCT) could be performed at the operator's discretion.

Following PCI, continuation of Aspirin in combination with either Clopidogrel, Prasugrel, or Ticagrelor was recommended, in line with ESC guidelines (23).

- Day 1 and hospitalization

During hospitalization, standard assessments were conducted according to local practice, including physical examination, vital signs, laboratory testing, and review of medications. Particular emphasis was placed on endpoints related to periprocedural myocardial infarction.

- A 12-lead ECG was to be performed within 24 hours post-procedure.
- Troponin levels and routine blood tests were to be obtained at least 6 hours, and no later than 48 hours, after PCI (20).
- In cases of multiple measurements, the highest troponin value recorded within 12–48 hours was to be reported.

- Discharge

After discharge, antithrombotic and/or anticoagulant therapy was to be continued according to guideline recommendations (ACCF/AHA/SCAI (22), ESC/EACTS (23)). Management of other concomitant medications, including oral anticoagulants (OACs), was left to the treating physician.

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The investigator was responsible for ensuring that all assessments related to primary and secondary endpoints were completed and documented in the eCRF (see Section 5.1).

- 1-month and 12 months follow-up visits (hospital visit or phone call)

At 1-month and 12-month follow-up visits (conducted either at the hospital or by phone), the following assessments were required, in line with routine clinical practice:

- Evaluation of Target Lesion Failure (TLF),
- Assessment of angina status,
- Review of medical history,
- Physical examination (if visit performed on-site),
- Review of concomitant medication use,
- Documentation of any endpoint-related events,
- Documentation of any protocol deviations.

Table 3. schedule of Assessment

Parameter/Examination	**Baseline**	**Day1 / Discharge**	**1 months** ± 7 days	**12 months** ± 1 month
Inclusion/exclusion	**X**			
Medical history	**X**		**X**	**X**
Angina status	**X**		**X**	**X**
Informed consent	**X**			
12-lead Electrocardiogram (ECG)	**X**	**X**		
PCI procedure	**X**			
Troponin level	**X** (Highly recommended: < 24h pre-procedure)	**X** (Between 6h and 48h post-procedure)		
Medication	**X**	**X**	**X**	**X**

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Adverse event monitoring		X	X	X

5.4.3 Core Laboratory evaluation

All procedures were independently reviewed by the Core Laboratory (CERC, Massy, France). An exhaustive qualitative and quantitative coronary angiographic analysis was performed using the Medis bifurcation QCA software (Version, Leiden, The Netherlands), including a detailed assessment of coronary lesions, PCI procedures, and results. Each analysis was independently reviewed by two experienced observers, both blinded to treatment allocation. In case of discrepancies, a third expert opinion was obtained.

QCA analyses were also performed for cases of target vessel revascularization, when procedural images were available. Further details are provided in the Core Laboratory Plan, version 1.0 dated September 7, 2020 (Appendix 2).

5.5 Ethical considerations

This study was conducted in compliance with the latest version of the Declaration of Helsinki, ICH-GCP and ISO 14155. Furthermore, it adhered to applicable laws and regulations of the countries where the study took place, including data protection legislation. The clinical investigational plan, Informed Consent Form (ICF), any other specific study documents and all amendments to these study documents were reviewed by the appropriate Ethics Committees (EC).

All documents, both initial and amended, were implemented only after obtaining the required approval or non-objection from the relevant EC(s).

Written Informed Consent was obtained from participants before any study specific screening procedures or baseline tests took place. Adverse events were reported to EC(s) and CA(s) in accordance with local regulation.

The Principal Investigator assured that no planned deviation from the CIP took place except where necessary to eliminate an immediate hazard to the study participants. The Principal Investigator promptly reported to the EC and the Sponsor any changes in research activity and all unanticipated problems involving risk to human participants, or others.

Patient confidentiality was maintained throughout the clinical study in a way that assured that data could always be tracked back to the source data. No patient

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identifiers were collected in any database and where applicable patient identifiers were removed (e.g. on source documents for corelab). For this purpose, a unique subject identification code (site ID – patient ID) was used that allowed identification of all data reported for each patient.

Financial disclose forms were not collected because data obtained from this study were not intended to be submitted to the FDA.

5.6 Data quality assurance

CERIC had the responsibility and liability for the initiation or implementation of the clinical investigation.

CERC was contracted by CERIC to perform some of the Sponsor's clinical investigation-related duties and functions, as specified in section 2.3, both sharing some regulatory responsibilities as contractually defined.

For the quality assurance of clinical investigation data, the Clinical Investigational Plan, CERC Standard Operating Procedures were followed. The following documents related to data quality were implemented:

- ✓ A study specific Clinical Monitoring Plan, detailing the study-specific monitoring strategy, requirements, and procedures involved in monitoring of the study; Appendix 4
- ✓ Study specific training plan to ensure clinical study personnel were adequately qualified to perform their study-specific assigned roles and responsibilities; (Appendix 5, of the quality plan).
- ✓ The Data Management Plan outlined procedures for data collection, processing, storage, and error resolution. Appendix 6

Data was collected through an electronic data capturing (EDC) system provided by Castor to the centers prior to study start.

Each site was contacted by the CRA after enrollment of the first two patients to ensure compliance with the protocol and the informed consent process.

A centralized monitoring (remote evaluation, in real-time and across sites of the electronic study data, performed at a location other than the sites) of this study was conducted by CERC to identify clinical investigation sites, with higher frequency of errors, protocol deviations or data anomalies, as well as to appraise site performance metrics. Electronic critical data outlined by the centralized monitoring were routinely reviewed to identify missing data, inconsistent data, data outliers, and

potential protocol deviations. This review was populated in a Data Management report which was sent to Clinical Research Associate (CRA) once per month to be used as a support for site oversight.

The on-site monitoring activities for this study were conducted by CERC in all centers to ensure that Investigators obligations were fulfilled including the following:

- the rights and well-being of patients were protected,
- the CIP and all applicable amendments were followed,
- the data recorded were accurately represented according to source document verification.

A total of three monitoring visits per site were expected. These visits were scheduled in accordance with the Clinical Monitoring Plan v2.0, dated June 13th, 2020 (Appendix 4), and are summarized as follows:

- **First Monitoring Visit (MV1):** Conducted after the inclusion of 15 patients.
- **Second Monitoring Visit (MV2):** Conducted after enrolment was closed.
- **Third Monitoring Visit (MV3):** Conducted once all patients had completed the 12-month follow-up.
-

The dates of all monitoring visits, stratified by site, are summarized in **Table 4.**

Country	Site number	Site Name	MV1	MV2	MV3
France	250-01	Institut Jacques Cartier	13th April 2021	13th Dec 2022	08th Jan 2024
France	250-02	CHU Toulouse	08th April 2021	5th Jan 2023	12th Jan 2024
France	250-03	Hôpital Privé Claude Galien	03rd Sep 2021	09th Dec2022	04th Jan 2024
France	250-04	Hôpital Privé Dijon Bourgogne	22th Sep 2021	10th Jan 2022	28th Nov 2023
France	250-05	CHU Caen	08th Sep 2021	14th Dec 2022	27-28th Nov 2023
France	250-06	CHU La Timone Marseille	14th Jan 2021	10th Jan 2022	11-12th Dec 2023
France	250-07	Clinique du Pont de Chaume	25th Jan 2022	06th Dec 2022	14-15th Nov 2023
France	250-08	CHU de Nîmes	22th Nov 2021	17-18th Jan 2023	6-7th Nov 2023
France	250-09	Clinique Saint-Hilaire	07th July 2021	28th Dec 2022	10th Jan 2024

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Italy	380-01	Policlinico Gemelli	NA	15th Dec 2022	12th Jan 2024
Italy	380-02	IRCCS Istituto Clinico Humanitas	28th June 2022	21-22th Dec2022	14th Dec 2023
Italy	380-03	IRCCS Policlinico San Donato	19th Dec 2022	09th Feb 2023	17th Jan 2024
Switzerland	756-01	Centre Hospitalier Universitaire Vaudois	NA	15th Dec 2022	10th Jan 2024
UK	826-01	Freeman Hospital Newcastle	NA	22th Dec 2023	10th Oct 2023
UK	826-02	Brighton Sussex Hospital	NA	23th Jan 2023	30th Nov 2023
Spain	724-01	Hospital Universitario Reina Sofia	20th July 2022	22th Nov 2022	05th Feb 2024
Spain	724-02	Hospital 12 de Octubre	NA	23th Nov 2022	22th Jan 2024
Spain	724-03	Hospital Clinico San Carlos	NA	24th Nov 2022	05th Feb 2024
Spain	724-04	Hospital de Cabueñes	NA	29th Nov 2022	29th Jan 2024
Portugal	620-01	Hospital de Santa Cruz	NA	13th Jan 2023	22th Jan 2023

NA in the MV1 column means that the site did not reach the target to schedule the first monitoring visit before the end of recruitment.

Clinical Research Associates were required direct access to subject medical records, study management documents, regulatory documents, and patient informed consent documents, as well as other potential applicable records not listed here.

During monitoring visits, the following was performed:

- 100% of patients informed consent
- 10% of the patients at each site (with a minimum of 2 per site) 100% Source Data Verified (SDV)
- 100% of endpoints-related events SDV (if not done remotely)

Between two site visits, the CRA maintained close follow-up with the sites to ensure proper research conduct, facilitate communication with the study staff, and prepare for upcoming monitoring visits.

To achieve this, the CRA contacted sites once per month via phone and/or email.

Site follow-up activities included:

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- Addressing issues or pending actions identified through centralized monitoring and providing support for their resolution,
- Requesting missing source documents for endpoint-related events to support SDV and CEC activities,
- Encouraging site staff to maintain enrollment momentum and ensure timely completion of the EDC and event reporting,
- Requesting missing essential study documents, such as (but not limited to) EC submissions/approvals/communications, completed training records, updated CVs, and copies of the Delegation of Responsibilities/Training Log in cases of staff updates,
- Communicating study timelines to ensure alignment,
- Providing additional training as deemed necessary.

Further details are mentioned in the study Clinical Monitoring Plan in Appendix 4.

5.7 Statistical design, analysis and justifications including

SAS Software 9.4 (SAS Institute Inc., Cary, NC, USA) were used to conduct data processing and statistical analyses.

5.7.1 Sample size calculation

Assuming a 15% rate of Myocardial Infarction (PPMI) as defined by ARC-2 criteria in the control group, accounting for a 6% dropout rate, and using an absolute non-inferiority margin of 7.5%, it was calculated that randomizing 596 subjects equally between the two arms would provide 80% power to demonstrate non-inferiority at a one-sided significance level of 0.05.

5.7.2 Study Population definition

Three analysis populations were defined as follows:

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Intention-to-treat set (ITT): The ITT population comprised all randomized participants. Analyses were performed according to the randomized treatment arm, regardless of any cross-over or deviations from the treatment protocol.

Per Protocol set (PP): The PP population was defined as a subset of the ITT population, excluding participants with major protocol deviations. Analyses in the PP population were performed according to the treatment arm to which participants were originally allocated.

The PP population was reviewed and finalized during the Blinded Data Review Meeting (BDRM), which was held after completion and cleaning of all study data and prior to database lock. All major protocol deviations leading to exclusion from the PP population were documented in the Analysis Sets Specification.

As-treated set (AT): The AT population included all randomized participants who underwent the procedure. Analyses in the AT population were performed according to the treatment actually received, rather than the randomized allocation.

The primary analyses were performed on the ITT and PP set. The AT was used for sensitivity analyses.

5.7.3 Statistical analysis methods

Categorical variables are presented as numbers and percentages. Continuous variables are represented as mean ± standard deviation or median and interquartile range depending on their distribution.

The difference between the two arms regarding the primary endpoint was analyzed using a Farrington-Manning Score Test.

Note that the analysis of the primary endpoint deviates from the suggested analysis in the clinical investigation plan.

The difference between the two treatment arms with respect to the primary endpoint was analyzed using a Farrington–Manning Score Test. It should be noted that this analysis deviated from the approach originally suggested in the clinical investigation plan. Further details are provided in the Statistical Analysis Plan (SAP), final version 2.0 dated March 15, 2023 (Appendix 7).

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Comparisons between the two arms for TLF, main branch TLR, side branch TLR, and definite or probable stent thrombosis were performed as follows: event rates were estimated from cumulative incidence functions. For TLF, non-cardiovascular mortality was considered a competing risk. For main branch TLR, side branch TLR, and definite or probable stent thrombosis, both cardiovascular and non-cardiovascular mortality were treated as competing risks. Between-group comparisons were presented as risk differences with two-sided 95% confidence intervals (CIs) and a Z-test. These analyses were conducted at both the 1-month and 12-month follow-up time points.

Differences between treatment arms regarding the proportion of participants with improvement in angina status from baseline to 1 and 12 months, technical success, and the need for cross-over were analyzed using a chi-square test, or Fisher's exact test when expected cell counts were fewer than five participants.

Differences in continuous parameters, including total procedure time, acute gain in the main and side branches, fluoroscopy time, total radiation dose, and total contrast dose, were analyzed using a two-sample t-test. In cases of serious deviations from normality, a Wilcoxon rank-sum test was applied. Results were expressed as the mean difference, or the Hodges–Lehmann estimate of the median difference, between treatment groups, with corresponding two-sided 95% CIs.

5.7.4 Subgroup analyses

Subgroup analyses were conducted for the primary endpoint and for TLF within the predefined subgroups listed below. For each subgroup level, the Farrington–Manning Score Test was applied to assess non-inferiority. In addition, tests for interaction between treatment group and subgroup were performed using a logistic regression model.

- Sex (Male, Female)
- Age (<70, ≥ 70)
- Target vessel type (LAD vs non-LAD)
- Medina class (XX1 vs XX0)
- Angiographic diameter of side branch (<2.25 vs ≥2.25 by corelab)
- Side branch residual stenosis (< 50% versus ≥ 50% by corelab QCA)

Within the SB intervention arm, additional analyses were performed to compare SB ballooning vs Kissing Balloon Technique.

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5.8 Clinical Investigational Plan amendments

Amendment #	Related CIP version	Amendment rational
1	V3.0	Prolongation of recruitment period

5.9 Specific committees

5.9.1 Clinical Events Committee

The CEC was managed by CERC. The CEC was made up of 3 interventional cardiologists. These physicians were not participants in the clinical study, had no conflict of interest regarding the device under evaluation and were independent from the clinical sites and the grant giver.

The CEC was responsible for adjudicating endpoint related events reported during the study, following established explicit rules in the CEC charter (Appendix 8) which outlines the data required and the process to be followed in order to classify a clinical event. The study definitions were provided to assist the CEC members in the accurate adjudication and classification of events.

All members of the CEC were blinded to the per patient randomization arm (as far as possible, when the Cath lab report was not necessary to adjudication the event).

Further details are mentioned in CEC Charter, final version 1.0 dated October 14th 2021, available in Appendix 8.

7 CEC meetings took place from 18th of March 2022 to 15th of January 2024. During the meetings 157 events, including the suspected peri procedural MI (raise of troponin) were reviewed by the CEC members.

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6 Study Results

6.1 Study schedule

First Patient First Visit	October 9th, 2020
Last Patient First Visit	December 8th, 2022
Last Patient Last Visit	December 18th, 2023

6.2 Patients' enrollment

A total of 617 patients were randomized in the study. 20 additional patients, compared with the initial sample size, were included due to unreported data to assess the primary endpoints for about 20 patients. Follow up rate was 99.5% at 30 days Fup and 98.7% at 12 months Fup **(Figure 2).**

Figure 2. Study Flowchart

The distribution of patients by country and by site is given in **Table 5**.

Country	Site	Principal Investigator	Enrolled Patients
France	Hôpital Privé Jacques Cartier	Dr. Antoinette Neylon	66
France	CHU Toulouse Rangueil	Prof. Thibault Lhermusier	76
France	Hôpital Privé Claude Galien	Dr. Stephane Champagne	47
France	Hôpital Privé Dijon Bourgogne	Dr. Philippe Brunel	46
France	CHU de Caen	Dr. Katrien Blanchart	38
France	CHU Timone, Marseille	Prof. Thomas Cuisset	37
France	Clinique du pont de chaume	Dr. Laurent Delorme	48
France	CHU Nîmes	Dr. Luc Cornillet	100
France	Clinique Saint-Hilaire	Dr. Matthieu Godin	31
Portugal	Centro Hospitalar de Lisboa Ocidental, Lisbon	Prof. Pedro Gonçalves	8
Spain	Hospital Universitario Reina Sofia	Dr. Manuel Pan	16
Spain	Hospital Universitario 12 de Octubre	Dr. Maria Teresa Velázquez Martín	16
Spain	Hospital Clinico San Carlos	Dr. Antonio Fernández Ortiz	4
Spain	Hospital de Cabuenes	Dr. Juan Rondan	14
Switzerland	CHUV Centre Hospitalier Universitaire Vaudois	Dr. Vladimir Rubimbura	6
Italy	Universita' Cattolica del Sacro CuoreLargo Agostino Gemelli	Dr. Francesco Burzotta	17
Italy	Istituto Clinico Humanitas	Dr. Bernhard Reimers	35
Italy	IRCCS Policlinico San Donato	Dr. Luca Testa	7
United Kingdom	Freeman Hospital Newcastle	Prof. Mohaned Egred	1
United Kingdom	Royal Sussex County Hospital	Dr. David Hildick-Smith	4

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6.3 CIP Compliance and patients dispositions

Patients with Protocol deviation (PD) were identified and classified (*e.g*. minor or major) prior to database lock and PP analysis. Major protocol deviations were identified for 76 patients who were excluded from PP population. One patient was randomized by mistake and was therefore excluded from the ITT population. The different categories of major protocol deviation are listed in the table 6 below, along with the number of occurrence per category of major PD.

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Table 6. Protocol Compliance

Patient Characteristic	Statistic	No side branch intervention	Side branch intervention	Total
Randomised	N	304	313	617
Included in Intention to Treat (ITT)	n/N (%)	303/304 (99.67%)	313/313 (100.00%)	616/617 (99.84%)
Included in Per Protocol Population (PP)	n/N (%)	268/303 (88.45%)	272/313 (86.90%)	540/616 (87.66%)
Included in As Treated Population (AT)	N	302	314	616
Crossover	N	6	5	11
Excluded from PP	n/N (%)	35/303 (11.55%)	41/313 (13.10%)	76/616 (12.34%)
Inclusion/Exclusion criteria not respected	n/N (%)	5/303 (1.65%)	2/313 (0.64%)	7/616 (1.14%)
Need to cross over	n/N (%)	6/303 (1.98%)	3/313 (0.96%)	9/616 (1.46%)
Main branch stenosis should be > 70%	n/N (%)	2/303 (0.66%)	1/313 (0.32%)	3/616 (0.49%)
Cardiac markers not (properly) done	n/N (%)	30/303 (9.90%)	31/313 (9.90%)	61/616 (9.90%)
Side branch low flow associated with ischemia should be No	n/N (%)	2/303 (0.66%)	0/313 (0.00%)	2/616 (0.32%)
POT not performed after SB ballooning	n/N (%)	0/303 (0.00%)	4/313 (1.28%)	4/616 (0.65%)

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6.4 Analysis

In this section, statistical analyses are presented using the ITT population (N=616). PP, AT and side branch subgroup analyses are given in the statistical report (Appendix 9).

6.4.1 Subject demographics and medical history

The average age of the patients was 67.4 years old in the no SB intervention group and 68.0 years old in SB intervention group. 75.9% were male in no SB intervention group and 78.6% in SB intervention group.

At baseline, 7.6% of patients in the no-SB intervention group and 4.5% in the SB intervention group had a history of cerebrovascular disease. Additionally, 26.4% of patients in the no-SB intervention group and 20.4% in the SB intervention group had a family history of coronary disease. Previous coronary intervention (PCI) was reported in 28.0% of patients in the no-SB intervention group and 37.0% in the SB intervention group. Previous myocardial infarction (MI) was present in 13.5% of patients in the no-SB intervention group and 27.5% in the SB intervention group, while previous coronary artery bypass grafting (CABG) was noted in 0.7% of patients in the no-SB intervention group and 0.3% in the SB intervention group.

Regarding lifestyle and comorbidities, 22.1% of patients in the no-SB intervention group and 20.1% in the SB intervention group were smokers. Diabetes was reported in 27.0% of patients in the no-SB intervention group and 27.2% in the SB intervention group, while hypertension was present in 63.0% of patients in the no-SB intervention group and 61.0% in the SB intervention group. Hypercholesterolemia was observed in 52.8% of patients in the no-SB intervention group and 57.5% in the SB intervention group.

There were no significant differences in baseline demographic or clinical characteristics between the two groups, except for previous PCI and previous MI, which were significantly more frequent in the SB intervention group.

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Table 7. Baseline Demographics and Clinical Characteristics

Baseline Characteristics	Statistic	Randomized treatment		P-value
		No side branch intervention	Side branch intervention	
Gender				0.426
Male	n/N (%)	230/303 (75.91%)	246/313 (78.59%)	
Female	n/N (%)	73/303 (24.09%)	67/313 (21.41%)	
Age	N	303	313	0.4733
	Mean	67.4	68.0	
	Std	11.19	10.49	
BMI [kg/m²]	N	303	312	0.8908
	Mean	28.3	28.5	
	Std	14.91	19.84	
History of Cerebrovascular Disease	n/N (%)	23/303 (7.59%)	14/313 (4.47%)	0.103
Peripheral Vascular disease	n/N (%)	30/303 (9.90%)	44/313 (14.06%)	0.113
Previous Coronary Intervention	n/N (%)	85/303 (28.05%)	116/313 (37.06%)	0.017
Previous Myocardial Infarction	n/N (%)	41/303 (13.53%)	86/313 (27.48%)	<.001
Previous CABG	n/N (%)	2/303 (0.66%)	1/313 (0.32%)	0.618
History of Congestive Heart Failure	n/N (%)	14/303 (4.62%)	7/313 (2.24%)	0.103
History of Renal Insufficiency	n/N (%)	11/303 (3.63%)	19/313 (6.07%)	0.160
Any Risk Factors	n/N (%)	286/303 (94.39%)	287/313 (91.69%)	0.189
Current Smoker	n/N (%)	67/303 (22.11%)	63/313 (20.13%)	0.546

Baseline Characteristics	Statistic	Randomized treatment		
		No side branch intervention	Side branch intervention	P-value
Former Smoker	n/N (%)	101/303 (33.33%)	102/313 (32.59%)	0.844
Diabetes	n/N (%)	82/303 (27.06%)	85/313 (27.16%)	0.979
Hypertension (requiring treatment)	n/N (%)	191/303 (63.04%)	191/313 (61.02%)	0.607
Hypercholesterolemia (requiring treatment)	n/N (%)	160/303 (52.81%)	180/313 (57.51%)	0.241
Family history of coronary disease	n/N (%)	80/303 (26.40%)	64/313 (20.45%)	0.081

6.4.2 Pre-Procedural Characteristics

The pre-procedure angina status was not significantly different between the two groups: 53.6% of patients in the no-SB intervention group and 50.8% in the SB intervention group presented with silent ischemia. Additionally, the pre-procedure angina class was not significantly different between the two groups: 44.3% of patients in the no-SB intervention group and 44.1% in the SB intervention group presented with CCS Class 2 angina.

Regarding medication before the procedure, there were no significant differences between the groups except for aspirin use, which was slightly higher in the no-SB group, with 90.4% of patients taking aspirin compared to 85.0% in the SB intervention group. Additionally, 78.8% of patients were taking a P2Y12 inhibitor in the no-SB intervention group compared to 76.0% in the SB intervention group.

The pre-procedure ECG results were not significantly different between the two groups: 72.5% of patients in the no-SB intervention group and 72.1% in the SB intervention group presented with a sinus rhythm on ECG.

There was no significant difference between the groups regarding pre-procedural troponin levels. High-sensitivity troponin T (hs-T) levels were out of range and clinically significant in 38.4% of patients in the no-SB intervention group and in 34.1% of patients in the SB intervention group. High-sensitivity troponin I (hs-I) levels were out of range and clinically significant in 9.6% of patients in the no-SB intervention group and in 12.6% of patients in the SB intervention group.

Table 8. Pre-Procedural Characteristics

		Randomized treatment		
Pre-Procedural Characteristic	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Angina Status				
Pre-procedure Angina status				0.892
Silent Ischemia	n/N (%)	162/302 (53.64%)	159/313 (50.80%)	
Stable angina	n/N (%)	85/302 (28.15%)	93/313 (29.71%)	
Unstable Angina	n/N (%)	27/302 (8.94%)	28/313 (8.95%)	
NSTEMI	n/N (%)	28/302 (9.27%)	33/313 (10.54%)	
Pre-procedure Angina class				0.364
CSS 1	n/N (%)	34/106 (32.08%)	41/118 (34.75%)	
CSS 2	n/N (%)	47/106 (44.34%)	52/118 (44.07%)	
CSS 3	n/N (%)	22/106 (20.75%)	17/118 (14.41%)	
CSS 4	n/N (%)	3/106 (2.83%)	8/118 (6.78%)	
Medications Before Procedure				
Any routine medication taken pre-procedure	n/N (%)	297/303 (98.02%)	301/313 (96.17%)	0.172

		Randomized treatment		
Pre-Procedural Characteristic	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Aspirin	n/N (%)	273/302 (90.40%)	266/313 (84.98%)	0.041
P2/Y12 inhibitors	n/N (%)	238/302 (78.81%)	238/313 (76.04%)	0.412
Clopidogrel	n/N (%)	181/302 (59.93%)	182/313 (58.15%)	0.652
Ticagrelor	n/N (%)	44/302 (14.57%)	46/313 (14.70%)	0.964
Prasugrel	n/N (%)	13/302 (4.30%)	11/313 (3.51%)	0.613
Oral anticoagulation	n/N (%)	33/302 (10.93%)	36/313 (11.50%)	0.821
ECG				
Pre-procedure Cardiac rhythm				0.264
Sinusal	n/N (%)	216/298 (72.48%)	220/305 (72.13%)	
Atrial Fibrillation	n/N (%)	11/298 (3.69%)	10/305 (3.28%)	
Pacemaker	n/N (%)	3/298 (1.01%)	5/305 (1.64%)	
Right bundle branch block	n/N (%)	10/298 (3.36%)	24/305 (7.87%)	
Left bundle branch block	n/N (%)	16/298 (5.37%)	13/305 (4.26%)	
Q-waves	n/N (%)	10/298 (3.36%)	8/305 (2.62%)	
ST-T abnormalities	n/N (%)	32/298 (10.74%)	25/305 (8.20%)	
Cardiac Enzymes				
Troponine hsT [µg/L]				0.403
Within range	n/N (%)	56/138 (40.58%)	67/138 (48.55%)	
Out-of-range but not clinically significant	n/N (%)	29/138 (21.01%)	24/138 (17.39%)	

		Randomized treatment		
Pre-Procedural Characteristic	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Out-of-range and clinically significant	n/N (%)	53/138 (38.41%)	47/138 (34.06%)	
Troponine hsI [µg/L]				0.641
Within range	n/N (%)	132/157 (84.08%)	134/167 (80.24%)	
Out-of-range but not clinically significant	n/N (%)	10/157 (6.37%)	12/167 (7.19%)	
Out-of-range and clinically significant	n/N (%)	15/157 (9.55%)	21/167 (12.57%)	

Additional statistics characterizing the Pre-Procedural Characteristics are given in the statistical reports (Appendix 9, Table 4).

6.4.3 Procedural Characteristics

Study lesion and procedural characteristics were described as outlined in Table 9. Most of the study bifurcations were classified as Medina class X,X,0. Medina classes were equally distributed among the two groups. According to site assessments, the mean MB lesion length was 22.2 mm in the no-SB intervention group and 22.3 mm in the SB intervention group. The mean pre-procedure stenosis in the proximal MB was 68.7% in the no-SB intervention group and 70.3% in the SB intervention group.

The mean SB lesion length was 3.8 mm in the no-SB intervention group and 4.8 mm in the SB intervention group, while the mean pre-procedure stenosis in the proximal SB was 26.3% in the no-SB intervention group and 32.5% in the SB intervention group.

There was no significant difference between the groups regarding MB assessment, whereas a significant difference was observed for the SB.

In the SB intervention group, the need for crossover was observed in 1.6% of patients, while in the no-SB intervention group, the need for crossover was observed in 2% of patients.

For patients randomized to the SB intervention group, kissing balloon inflation was performed in 43% of patients to dilate the SB, whereas SB ballooning alone was used in 57% of patients. In the no-SB intervention group (crossover patients), kissing balloon inflation was performed in 66.7% of patients to dilate the SB, while SB ballooning alone was used in 33.3% of patients.

Procedural success was achieved in 100% of patients. Procedural complications occurred in less than 5% of patients, with no significant differences observed between the two groups.

Additional statistics characterizing the implantation procedures are given in the statistical reports (Appendix 9, Table 5).

Table 9. Procedural Charcteristics

		Randomized treatment		
Procedural Characteristic	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Total Number of Patients & Procedures	N	303	313	
Medina classification				0.229
1,1,1	n/N (%)	61/303 (20.13%)	77/313 (24.60%)	
1,1,0	n/N (%)	115/303 (37.95%)	93/313 (29.71%)	
1,0,1	n/N (%)	5/303 (1.65%)	10/313 (3.19%)	
1,0,0	n/N (%)	26/303 (8.58%)	25/313 (7.99%)	
0,1,1	n/N (%)	26/303 (8.58%)	34/313 (10.86%)	
0,1,0	n/N (%)	70/303 (23.10%)	74/313 (23.64%)	
Tortuosity				0.913

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		Randomized treatment		
Procedural Characteristic	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
None	n/N (%)	153/303 (50.50%)	156/313 (49.84%)	
Mild	n/N (%)	115/303 (37.95%)	115/313 (36.74%)	
Moderate	n/N (%)	27/303 (8.91%)	33/313 (10.54%)	
Severe	n/N (%)	8/303 (2.64%)	9/313 (2.88%)	
Main Branch Lesion Length [mm]	N	303	313	0.9933
	Mean	22.2	22.3	
	Std	8.81	9.24	
Main Branch (MB) TIMI Flow Pre-Procedure				0.700
2	n/N (%)	5/303 (1.65%)	4/313 (1.28%)	
3	n/N (%)	298/303 (98.35%)	309/313 (98.72%)	
Proximal MB Reference Vessel Diameter [mm]	N	303	313	0.1076
	Mean	3.37	3.29	
	Std	0.519	0.475	
Proximal MB Stenosis Pre-Procedure [%]	N	303	313	0.5899
	Mean	68.7	70.3	
	Std	31.98	31.72	
Distal MB Reference Vessel Diameter [mm]	N	303	312	0.0917
	Mean	2.92	2.87	
	Std	0.407	0.410	

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		Randomized treatment		
Procedural Characteristics	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Distal MB Stenosis Pre-Procedure [%]	N	303	313	0.2719
	Mean	74.7	72.7	
	Std	25.00	26.50	
Side Branch (MB) TIMI Flow Pre-Procedure				0.582
2	n/N (%)	1/303 (0.33%)	2/313 (0.64%)	
3	n/N (%)	302/303 (99.67%)	311/313 (99.36%)	
Side Branch Lesion Length [mm]	N	302	313	0.0186
	Mean	3.8	4.8	
	Std	5.67	5.85	
Side Branch Reference Vessel Diameter [mm]	N	303	313	0.0337
	Mean	2.53	2.50	
	Std	0.307	0.282	
Side Branch Stenosis Pre-Procedure [%]	N	303	313	0.0286
	Mean	26.3	32.5	
	Std	31.35	34.49	
Preparation of Main Branch				
Balloon	n/N (%)	255/303 (84.16%)	256/313 (81.79%)	0.434
Cutting Balloon	n/N (%)	6/303 (1.98%)	3/313 (0.96%)	0.291

Procedural Characteristics	Statistic	Randomized treatment		
		No side branch intervention	Side branch intervention	P-value
Roblator	n/N (%)	4/303 (1.32%)	4/313 (1.28%)	0.963
Shock Wave	n/N (%)	1/303 (0.33%)	1/313 (0.32%)	0.982
Other	n/N (%)	3/303 (0.99%)	8/313 (2.56%)	0.142
Not Done	n/N (%)	43/303 (14.19%)	46/313 (14.70%)	0.859
Number of Study Stents				
Total Number of Study Stents Implanted in MB	Sum	347	355	
	[n] Mean (SD)	[303] 1.1 (0.4)	[313] 1.1 (0.4)	
Total Number of Study Stents Implanted in SB	Sum	1	14	
	[n] Mean (SD)	[1] 1.0 (.)	[14] 1.0 (0.0)	
POT (Proximal Optimization Technique)				
Maximum post-dilatation balloon diameter [mm]	N	298	311	0.0814
	Mean	3.4	3.4	
Maximum post-dilatation balloon length [mm]	N	300	313	0.4081
	Mean	10.5	10.9	
	Std	3.82	4.50	
Maximum pressure in Main Branch [atm]	N	299	311	0.0214
	Mean	16.2	15.6	
	Std	3.97	3.85	

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		Randomized treatment		
Procedural Characteristics	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Total Number of Patients & Procedures	N	303	313	
If Randomised to SB, Need to Cross-Over	n/N (%)	0/ 0	5/313 (1.60%)	
Failure to Wire the SB	n/N (%)		1/4 (25.00%)	
Failure to Place the Balloon in the SB	n/N (%)		1/4 (25.00%)	
Other	n/N (%)		2/4 (50.00%)	
If Randomised to no SB, Need to Cross-Over	n/N (%)	6/303 (1.98%)	0/ 0	
SB Occlusion	n/N (%)	1/303 (0.33%)	0/312 (0.00%)	
SB Dissection	n/N (%)	0/303 (0.00%)	0/312 (0.00%)	
SB Low Flow	n/N (%)	2/303 (0.66%)	0/312 (0.00%)	
Ischaemia	n/N (%)	3/303 (0.99%)	0/312 (0.00%)	
SB intervention Done	n/N (%)	6/303 (1.98%)	308/313 (98.40%)	
Side Branch Intervention				
Kissing Balloon	n/N (%)	4/6 (66.67%)	132/307 (43.00%)	
SB Ballooning	n/N (%)	2/6 (33.33%)	175/307 (57.00%)	
Kissing Balloon				
Inflation Successful	n/N (%)	3/4 (75.00%)	127/132 (96.21%)	
MB: Balloons Used for Kissing Inflation				
Compliant	n/N (%)	1/3 (33.33%)	51/123 (41.46%)	
Non-compliant	n/N (%)	2/3 (66.67%)	72/123 (58.54%)	

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Procedural Characteristics	Statistic	Randomized treatment		P-value
		No side branch intervention	Side branch intervention	
Maximum Kissing Balloon Diameter [mm] (MB)	[n] Mean (SD)	[3] 3 (0)	[127] 3 (0)	
Maximum Kissing Balloon Length [mm] (MB)	[n] Mean (SD)	[3] 14 (2)	[127] 13 (2)	
Maximum Kissing Balloon pressure in MB [atm]	[n] Mean (SD)	[2] 13 (4)	[127] 12 (3)	
SB: Balloons Used for Kissing Inflation				
Compliant	n/N (%)	1/3 (33.33%)	58/123 (47.15%)	
Non-compliant	n/N (%)	2/3 (66.67%)	65/123 (52.85%)	
Maximum Kissing Balloon Diameter [mm] (SB)	[n] Mean (SD)	[3] 3 (0)	[127] 2 (0)	
Maximum Kissing Balloon Length [mm] (SB)	[n] Mean (SD)	[3] 12 (3)	[127] 13 (2)	
Maximum Kissing Balloon pressure in SB [atm]	[n] Mean (SD)	[2] 12 (3)	[127] 11 (3)	
SB Ballooning				
Inflation Successful	n/N (%)	2/ 2 (100.00%)	167/175 (95.43%)	
Balloons Used for SB Ballooning				
Compliant	n/N (%)	2/2 (100.00%)	108/166 (65.06%)	
Non-compliant	n/N (%)	0/2 (0.00%)	58/166 (34.94%)	
Maximum SB Balloon Diameter [mm] (MB)	[n] Mean (SD)	[2] 2 (0)	[167] 3 (2)	
Maximum SB Balloon Length [mm] (MB)	[n] Mean (SD)	[2] 12 (0)	[167] 12 (3)	
Maximum Balloon pressure in SB [atm]	[n] Mean (SD)	[2] 10 (3)	[167] 13 (3)	
Further Treatment to SB Needed	n/N (%)	1/ 6 (16.67%)	18/308 (5.84%)	
Stent Implanted	n/N (%)	1/ 1 (100.00%)	14/ 18 (77.78%)	

Procedural Characteristics	Statistic	Randomized treatment		
		No side branch intervention	Side branch intervention	P-value
Technique Used				
T-Stenting	n/N (%)	1/1 (100.00%)	4/14 (28.57%)	
Tap	n/N (%)	0/1 (0.00%)	9/14 (64.29%)	
Other	n/N (%)	0/1 (0.00%)	1/14 (7.14%)	
Non-Study Stent Implanted	n/N (%)	0/ 1 (0.00%)	1/ 14 (7.14%)	
Side Branch Occlusion	n/N (%)	0/ 6 (0.00%)	1/308 (0.32%)	
Side branch low flow associated with ischemia	n/N (%)	0/ 6 (0.00%)	1/308 (0.32%)	
Main Branch Post TIMI Flow				
2	n/N (%)	0/303 (0.00%)	1/312 (0.32%)	
3	n/N (%)	303/303 (100.00%)	311/312 (99.68%)	
Main Branch Stenosis Post-Procedure [%]	N	303	310	0.3400
	Mean	2.1	2.4	
	Std	11.35	11.05	
Side Branch Post TIMI Flow				0.615
1	n/N (%)	0/303 (0.00%)	1/312 (0.32%)	
2	n/N (%)	1/303 (0.33%)	1/312 (0.32%)	
3	n/N (%)	302/303 (99.67%)	310/312 (99.36%)	

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Side Branch Stenosis Post-Procedure [%]	N	303	310	0.0010
	Mean	28.2	18.2	
	Std	33.00	26.40	
	(Q1, Q3)	(0.0; 50.0)	(0.0; 30.0)	
	Range	(0.0; 99.0)	(0.0; 99.0)	
Total Number of Patients & Procedures	N	303	313	
Procedural Success	n/N (%)	303/303 (100.00%)	313/313 (100.00%)	
Complete Revascularisation	n/N (%)	283/303 (93.40%)	285/313 (91.05%)	0.278
Non-complete Revascularisation: therapeutic decision				0.809
Medical treatment	n/N (%)	7/20 (35.00%)	12/28 (42.86%)	
Staged procedure	n/N (%)	4/20 (20.00%)	4/28 (14.29%)	
None	n/N (%)	9/20 (45.00%)	12/28 (42.86%)	
Procedural complication	n/N (%)	10/303 (3.30%)	15/313 (4.79%)	0.416
Main Branch	n/N (%)	4/302 (1.32%)	4/313 (1.28%)	1.000
Coronay Dissection	n/N (%)	4/303 (1.32%)	3/313 (0.96%)	0.721
Perforation	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Decrease Of Timi Flow <3	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Acute Occlusion	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Thrombus Formation	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Significant Air Embolism	n/N (%)	0/303 (0.00%)	1/313 (0.32%)	1.000

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Severe Arrhythmia	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Other	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Side Branch	n/N (%)	5/302 (1.66%)	10/313 (3.19%)	0.297
Coronay Dissection	n/N (%)	0/303 (0.00%)	9/313 (2.88%)	0.004
Perforation	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Decrease Of Timi Flow <3	n/N (%)	1/303 (0.33%)	0/313 (0.00%)	0.492
Acute Occlusion	n/N (%)	3/303 (0.99%)	0/313 (0.00%)	0.118
Thrombus Formation	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Significant Air Embolism	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Severe Arrhythmia	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	
Other	n/N (%)	2/303 (0.66%)	1/313 (0.32%)	0.618
Main Branch Complication				
Resolved	n/N (%)	3/ 4 (75.00%)	4/ 4 (100.00%)	1.000
Resolved With Sequelae	n/N (%)	1/ 4 (25.00%)	0/ 4 (0.00%)	1.000
Ongoing	n/N (%)	0/ 4 (0.00%)	0/ 4 (0.00%)	
Resulted In Death	n/N (%)	0/ 4 (0.00%)	0/ 4 (0.00%)	
Side Branch Complication				
Resolved	n/N (%)	5/ 5 (100.00%)	9/ 10 (90.00%)	1.000
Resolved With Sequelae	n/N (%)	0/ 5 (0.00%)	0/ 10 (0.00%)	
Ongoing	n/N (%)	0/ 5 (0.00%)	1/ 10 (10.00%)	1.000
Resulted In Death	n/N (%)	0/ 5 (0.00%)	0/ 10 (0.00%)	
Global outcome : death	n/N (%)	0/303 (0.00%)	0/313 (0.00%)	

6.4.4 Corelab Analysis

Corelab analysis did not confirm the significant difference in SB assessment between the two groups as reported by site. Lesion length and stenosis, as assessed by the corelab, were not found to be significantly different between the two groups.

As expected, in the SB, Corelab analysis revealed significant differences in acute gain and post-procedure % stenosis, with a higher acute gain and a lower % stenosis observed in the SB intervention group.

Further statistics describing Corelab Analysis are provided in the statistical reports. (Appendix 9, Table 12).

Table 10. Baseline QCA Statistical - Bifurcation analysis in lesion

		Randomized treatment		
QCA In Lesion	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Total Number of Patients analysed	N	303	313	
Total Number of Lesions analysed	N	303	317	
Stenosis localisation in lesion (% diameter stenosis)				
Proximal main	[n] Mean (SD)	[301] 45.06 (22.54)	[312] 44.26 (22.34)	0.658
Distal main	[n] Mean (SD)	[300] 53.12 (17.12)	[312] 50.99 (17.21)	0.125
Side branch	[n] Mean (SD)	[301] 32.11 (17.76)	[311] 32.60 (18.25)	0.741
Stenosis localisation in lesion (% stenosis >=50%)				
Proximal main	n/N (%)	143/301 (47.51%)	134/312 (42.95%)	0.2587
Distal main	n/N (%)	192/300 (64.00%)	188/312 (60.26%)	0.3415
Side branch	n/N (%)	57/301 (18.94%)	66/311 (21.22%)	0.4807

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		Randomized treatment		
QCA In Lesion	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Max stenosis in lesion (Segm 2 + 3 + 5) PRE				
MLD (mm)	[n] Mean (SD)	[301] 0.88 (0.30)	[312] 0.88 (0.28)	0.884
	Median (Q1; Q3)	0.87 (0.67; 1.08)	0.86 (0.70; 1.04)	
	Range	(0.00; 2.15)	(0.25; 1.88)	
% Diam Stenosis (%)	[n] Mean (SD)	[301] 63.19 (11.77)	[312] 62.34 (11.15)	0.360
	Median (Q1; Q3)	62.56 (55.11; 71.39)	62.65 (54.69; 70.83)	
	Range	(28.06; 100.00)	(20.04; 88.25)	
Main Branch (Segm 2 + 3)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 0.94 (0.35)	[312] 0.96 (0.34)	0.594
MLD (mm) - POST	[n] Mean (SD)	[301] 2.23 (0.43)	[312] 2.19 (0.43)	0.294
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 62.38 (12.62)	[312] 61.24 (12.29)	0.258
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 15.86 (8.54)	[312] 15.98 (7.79)	0.862
Acute Gain (mm)	[n] Mean (SD)	[301] 1.29 (0.49)	[310] 1.24 (0.46)	0.223
Prox Main Branch (Segm 2)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 1.68 (0.74)	[312] 1.66 (0.71)	0.755
MLD (mm) - POST	[n] Mean (SD)	[301] 2.77 (0.44)	[312] 2.80 (0.43)	0.419
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 45.06 (22.54)	[312] 44.26 (22.34)	0.658
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 12.05 (8.35)	[312] 11.02 (8.22)	0.127

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Reference diameter (mm) - PRE	[n] Mean (SD)	[301] 3.07 (0.53)	[312] 3.02 (0.57)	0.248
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 3.16 (0.50)	[312] 3.16 (0.50)	0.928
Obstruction length pre (mm)	[n] Mean (SD)	[303] 8.90 (6.20)	[314] 8.92 (6.46)	0.963
Acute Gain (mm)	[n] Mean (SD)	[301] 1.09 (0.69)	[310] 1.14 (0.68)	0.387
Distal Main Branch (Segm 3)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 1.07 (0.44)	[312] 1.11 (0.49)	0.333
MLD (mm) - POST	[n] Mean (SD)	[301] 2.24 (0.44)	[312] 2.20 (0.44)	0.302
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[300] 53.12 (17.12)	[312] 50.99 (17.21)	0.125
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 11.45 (8.50)	[312] 12.10 (8.74)	0.349
Reference diameter (mm) - PRE	[n] Mean (SD)	[300] 2.30 (0.42)	[312] 2.24 (0.46)	0.101
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 2.52 (0.40)	[312] 2.50 (0.40)	0.495
Obstruction length pre (mm)	[n] Mean (SD)	[303] 10.74 (7.46)	[314] 11.22 (7.87)	0.437
Acute Gain (mm)	[n] Mean (SD)	[301] 1.17 (0.54)	[310] 1.10 (0.49)	0.101
Side Branch (Segm 5)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 1.37 (0.45)	[312] 1.29 (0.43)	0.041
MLD (mm) - POST	[n] Mean (SD)	[301] 1.33 (0.49)	[312] 1.40 (0.44)	0.081
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 32.11 (17.76)	[311] 32.60 (18.25)	0.741
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 34.57 (18.70)	[312] 27.87 (16.82)	<.001
Reference diameter (mm) - PRE	[n] Mean (SD)	[300] 2.01 (0.38)	[311] 1.93 (0.40)	0.016
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 2.03 (0.39)	[312] 1.94 (0.41)	0.006
Obstruction length pre (mm)	[n] Mean (SD)	[303] 4.43 (3.04)	[314] 4.85 (3.65)	0.121
Acute Gain (mm)	[n] Mean (SD)	[301] -0.04 (0.36)	[310] 0.10 (0.41)	<.001

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Table 11. Corelab Baseline QCA Statistical - Bifurcation analysis in segment

		Randomized treatment		
QCA in segment	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Total Number of Patients analysed	N	303	313	
Total Number of Lesions analysed	N	303	317	
Stenosis localisation in segment (% diameter stenosis)				
Proximal edge	[n] Mean (SD)	[301] 9.58 (8.44)	[312] 10.48 (9.10)	0.204
Proximal main	[n] Mean (SD)	[301] 45.06 (22.54)	[312] 44.26 (22.34)	0.658
Distal main	[n] Mean (SD)	[300] 53.12 (17.12)	[312] 50.99 (17.21)	0.125
Distal edge	[n] Mean (SD)	[300] 10.43 (8.09)	[312] 9.20 (7.62)	0.055
Side branch	[n] Mean (SD)	[301] 32.11 (17.76)	[311] 32.60 (18.25)	0.741
	Median (Q1; Q3)	30.64 (18.57; 43.97)	32.15 (18.03; 47.56)	
	Range	(0.37; 100.00)	(-10.94; 79.40)	
Side branch edge	[n] Mean (SD)	[301] 11.04 (9.48)	[311] 11.78 (8.78)	0.312
	Median (Q1; Q3)	9.33 (5.24; 15.07)	9.90 (5.89; 16.04)	
	Range	(-6.67; 100.00)	(-4.35; 66.88)	
Stenosis localisation in lesion (% stenosis >=50%)				
Proximal edge	n/N (%)	1/301 (0.33%)	1/312 (0.32%)	0.2587
Proximal main	n/N (%)	143/301 (47.51%)	134/312 (42.95%)	0.3415
Distal main	n/N (%)	192/300 (64.00%)	188/312 (60.26%)	0.3169

Distal edge	n/N (%)	1/300 (0.33%)	0/312 (0.00%)	0.4807
Side branch	n/N (%)	57/301 (18.94%)	66/311 (21.22%)	0.5460
Side branch edge	n/N (%)	2/301 (0.66%)	1/311 (0.32%)	0.544
Max stenosis in lesion (Segm 1 to 6) PRE				
MLD (mm)	[n] Mean (SD)	[301] 0.88 (0.30)	[312] 0.88 (0.28)	0.869
% Diam Stenosis (%)	[n] Mean (SD)	[301] 63.19 (11.77)	[312] 62.35 (11.15)	0.363
Main Branch (Segm 1 to 4)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 0.94 (0.35)	[312] 0.96 (0.34)	0.598
MLD (mm) - POST	[n] Mean (SD)	[301] 1.99 (0.41)	[312] 1.95 (0.40)	0.207
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 62.38 (12.62)	[312] 61.26 (12.24)	0.262
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 21.07 (8.64)	[312] 21.80 (8.37)	0.286
Acute Gain (mm)	[n] Mean (SD)	[301] 1.05 (0.46)	[310] 1.00 (0.41)	0.142
Prox Main Branch (Segm 1+2)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 1.66 (0.72)	[312] 1.65 (0.69)	0.796
MLD (mm) - POST	[n] Mean (SD)	[301] 2.67 (0.45)	[312] 2.66 (0.45)	0.828
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 45.51 (21.86)	[312] 44.61 (21.79)	0.611
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 14.74 (7.53)	[312] 14.98 (7.64)	0.692
Reference diameter (mm) - PRE	[n] Mean (SD)	[301] 3.09 (0.52)	[312] 3.04 (0.56)	0.202
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 3.20 (0.51)	[312] 3.19 (0.51)	0.943
Obstruction length pre (mm)	[n] Mean (SD)	[303] 8.90 (6.20)	[314] 8.92 (6.46)	0.963
Acute Gain (mm)	[n] Mean (SD)	[301] 1.01 (0.69)	[310] 1.01 (0.70)	0.890
Distal Main Branch (Segm 3+4)				

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MLD (mm) - PRE	[n] Mean (SD)	[301] 1.07 (0.44)	[312] 1.10 (0.48)	0.330
MLD (mm) - POST	[n] Mean (SD)	[301] 2.00 (0.42)	[312] 1.95 (0.40)	0.186
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[300] 53.23 (16.88)	[312] 51.10 (16.95)	0.119
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 18.19 (9.06)	[312] 19.36 (8.73)	0.104
Reference diameter (mm) - PRE	[n] Mean (SD)	[300] 2.31 (0.42)	[312] 2.26 (0.45)	0.132
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 2.52 (0.40)	[312] 2.50 (0.40)	0.519
Obstruction length pre (mm)	[n] Mean (SD)	[303] 10.74 (7.46)	[314] 11.22 (7.87)	0.437
Acute Gain (mm)	[n] Mean (SD)	[301] 0.93 (0.52)	[310] 0.85 (0.46)	0.055
Side Branch (Segm 5+6)				
MLD (mm) - PRE	[n] Mean (SD)	[301] 1.34 (0.43)	[312] 1.26 (0.41)	0.015
MLD (mm) - POST	[n] Mean (SD)	[301] 1.30 (0.46)	[312] 1.33 (0.39)	0.316
% Diam Stenosis (%) - PRE	[n] Mean (SD)	[301] 32.85 (17.04)	[311] 33.76 (16.95)	0.505
% Diam Stenosis (%) - POST	[n] Mean (SD)	[301] 35.95 (17.44)	[312] 30.67 (15.09)	<.001
Reference diameter (mm) - PRE	[n] Mean (SD)	[300] 2.02 (0.38)	[311] 1.95 (0.39)	0.012
Reference diameter (mm) - POST	[n] Mean (SD)	[301] 2.05 (0.39)	[312] 1.96 (0.41)	0.005
Obstruction length pre (mm)	[n] Mean (SD)	[303] 4.43 (3.04)	[314] 4.85 (3.65)	0.121
Acute Gain (mm)	[n] Mean (SD)	[301] -0.04 (0.34)	[310] 0.07 (0.35)	<.001

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6.4.5 Post-procedure Characteristics

The post-procedure angina status was not significantly different between the two groups: 96.6% of patients in the no-SB intervention group and 97.4% in the SB intervention group were asymptomatic. Additionally, the post-procedure angina class was not significantly different between the two groups: 65.0% of patients in the no-SB intervention group and 76.5% in the SB intervention group presented with CCS Class 1 angina.

The post-procedure ECG results were not significantly different between the two groups: 72.4% of patients in the no-SB intervention group and 71.8% in the SB intervention group presented with a sinus rhythm on ECG, 4.38% of patients in the no-SB intervention group and 5.3% in the SB intervention group presented a new Q wave on ECG, and 9.76% of patients in the no-SB intervention group and 7.62% in the SB intervention group presented with a ST segment deviation.

High-sensitivity troponin T (hs-T) levels were out of range and clinically significant in 35.6% of patients in the no-SB intervention group and in 38.2% of patients in the SB intervention group. High-sensitivity troponin I (hs-I) levels were out of range and clinically significant in 18.7% of patients in the no-SB intervention group and in 24.5% of patients in the SB intervention group. There was no significant difference between the two groups.

Further statistics describing post-procedural characteristics are provided in the statistical reports. (Appendix 9, Table 6).

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Table 12. Post procedural Characteristics

Post-procedural Characteristics	Statistic	Randomized treatment		P-value
		No side branch intervention	Side branch intervention	
Total Number of Patients	N	303	313	
Angina Status				
Angina status				0.583
Asymptomatic	n/N (%)	288/298 (96.64%)	300/308 (97.40%)	
Angina	n/N (%)	10/298 (3.36%)	8/308 (2.60%)	
Angina Class				0.667
CSS 1	n/N (%)	13/20 (65.00%)	13/17 (76.47%)	
CSS 2	n/N (%)	4/20 (20.00%)	1/17 (5.88%)	
CSS 3	n/N (%)	1/20 (5.00%)	1/17 (5.88%)	
CSS 4	n/N (%)	2/20 (10.00%)	2/17 (11.76%)	
ECG				
Cardiac rhythm				0.824
Sinusal	n/N (%)	215/297 (72.39%)	217/302 (71.85%)	
Atrial Fibrillation	n/N (%)	4/297 (1.35%)	8/302 (2.65%)	
Pacemaker	n/N (%)	2/297 (0.67%)	3/302 (0.99%)	
Right bundle branch block	n/N (%)	16/297 (5.39%)	19/302 (6.29%)	
Left bundle branch block	n/N (%)	18/297 (6.06%)	16/302 (5.30%)	

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Post-procedural Characteristics	Statistic	Randomized treatment		P-value
		No side branch intervention	Side branch intervention	
Q-waves	n/N (%)	13/297 (4.38%)	16/302 (5.30%)	
ST-T abnormalities	n/N (%)	29/297 (9.76%)	23/302 (7.62%)	
Cardiac Enzymes				
Troponine hsT [µg/L]				0.182
Within range	n/N (%)	18/135 (13.33%)	9/136 (6.62%)	
Out-of-range but not clinically significant	n/N (%)	69/135 (51.11%)	75/136 (55.15%)	
Out-of-range and clinically significant	n/N (%)	48/135 (35.56%)	52/136 (38.24%)	
Troponine hsI [µg/L]				0.076
Within range	n/N (%)	34/155 (21.94%)	21/163 (12.88%)	
Out-of-range but not clinically significant	n/N (%)	92/155 (59.35%)	102/163 (62.58%)	
Out-of-range and clinically significant	n/N (%)	29/155 (18.71%)	40/163 (24.54%)	

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6.4.6 Antithrombotic regimen

Regarding medication during follow-up, there were no significant differences between the groups. At 1 month, more than 80% of patients were taking aspirin, and more than 90% were taking a P2Y12 inhibitor. At the 12-month follow-up, more than 75% of patients were taking aspirin, and more than 50% were taking a P2Y12 inhibitor.

Table 13. Antithrombotic regimen during follow-up

		Randomized treatment		
Antithrombotic regimen	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Total Number of Patients	N	303	313	
At 1 Month				
Aspirin	n/N (%)	250/300 (83.33%)	257/311 (82.64%)	0.830
P2/Y12 inhibitors	n/N (%)	278/300 (92.67%)	291/311 (93.57%)	0.750
Clopidogrel	n/N (%)	210/300 (70.00%)	220/311 (70.74%)	0.860
Ticagrelor	n/N (%)	50/300 (16.67%)	53/311 (17.04%)	0.914
Prasugrel	n/N (%)	20/300 (6.67%)	18/311 (5.79%)	0.738
Oral anticoagulation	n/N (%)	34/300 (11.33%)	38/311 (12.22%)	0.802
Aspirin + P2Y12	n/N (%)	237/246 (96.34%)	243/249 (97.59%)	0.445
Aspirin or P2Y12 + VKA or NOAC	n/N (%)	28/ 31 (90.32%)	35/ 38 (92.11%)	1.000
At 12 Months				

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		Randomized treatment		
Antithrombotic regimen	**Statistic**	**No side branch intervention**	**Side branch intervention**	**P-value**
Aspirin	n/N (%)	230/292 (78.77%)	232/305 (76.07%)	0.436
P2/Y12 inhibitors	n/N (%)	160/292 (54.79%)	172/305 (56.39%)	0.742
Clopidogrel	n/N (%)	124/292 (42.47%)	127/305 (41.64%)	0.868
Ticagrelor	n/N (%)	23/292 (7.88%)	34/305 (11.15%)	0.210
Prasugrel	n/N (%)	14/292 (4.79%)	11/305 (3.61%)	0.542
Oral anticoagulation	n/N (%)	35/292 (11.99%)	44/305 (14.43%)	0.400
Aspirin + P2Y12	n/N (%)	133/168 (79.17%)	139/179 (77.65%)	0.795
Aspirin or P2Y12 + VKA or NOAC	n/N (%)	23/ 46 (50.00%)	28/ 52 (53.85%)	0.840

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6.4.7 Primary Endpoint Analysis

Primary endpoint results are presented in Table 14 to Table 16. In both the ITT and PP populations, the rate of peri-procedural MI in the no-SB intervention group was significantly non-inferior to that in the SB intervention group but not superior. The PPMI rate was then compared within the SB intervention group between the kissing balloon and SB ballooning strategies. SB ballooning was non-inferior to kissing balloon treatment and slightly superior, though not significantly.

Table 14. Primary endpoint- ITT set

		Randomized treatment			
Primary Endpoint	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Risk Difference (95% CI)**	**P-value**
Total Number of Patients	N	303	313		
Peri-Procedural MI at Discharge or 48h Post-Procedure	n/N	11/273	16/282		
	% (95% CI)	4.1% (1.7; 6.4)	5.7% (2.9; 8.4)	-1.6% (-∞; 1.4)	<.0001$
				-1.6% (-5.2; 2.0)	0.3840#

n= number of events; N= number of patients in the analysis, $= one-sided non-inferiority test with a margin of 7.56%.
#: two-sided superiority test.
Missing data were accounted for by performing 100 multiple imputations.

Table 15. Primary endpoint- PP set

		Randomized treatment			
Primary Endpoint	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Risk Difference (95% CI)**	**P-value**
Total Number of Patients	N	268	272		
Peri-Procedural MI at Discharge or 48h Post-Procedure	n/N	11/266	16/269		
	% (95% CI)	4.1% (1.7; 6.5)	5.9% (3.1; 8.8)	-1.8% (-∞; 1.3)	<.0001$
				-1.8% (-5.5; 1.9)	0.3451#

n= number of events; N= number of patients in the analysis, $= one-sided non-inferiority test with a margin of 7.56%.
#: two-sided superiority test.
Missing data were accounted for by performing 100 multiple imputations.

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Table 16. Side Branch Intervention Group

			Type of SB technique		
Primary Endpoint	**Statistic**	**SB Ballooning**	**Kissing Ballooon**	**Risk Difference (95% CI)**	**P-value**
Total Number of Patients	N	175	132		
Peri-Procedural MI at Discharge or 48h Post-Procedure	n/N	5/156	11/120		
	% (95% CI)	3.4% (0.5; 6.4)	8.9% (3.8; 13.9)	-5.5% (-∞; -0.6)	<.0001$
				-5.5% (-11; 0.4)	0.0667#

n= number of events; N= number of patients in the analysis, $= one-sided non-inferiority test with a margin of 7.56%.
#: two-sided superiority test.
Missing data were accounted for by performing 100 multiple imputations.

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6.4.8 Secondary Analysis

Secondary procedural endpoint results are presented in Table 17. There was no significant difference between the SB and no-SB intervention groups with respect to technical success (either in the Main Branch or Side Branch) or the need for crossover. However, there was a significant difference between the groups regarding procedural time, fluoroscopy time, contrast volume, and radiation exposure.

Further statistics describing Secondary Procedural Endpoints are provided in the statistical reports. (Appendix 9, Table 9).

Table 17. Secondary Procedural Endpoints

		Randomized treatment			
Secondary Procedural Endpoints	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Difference (95% CI)**	**P-value**
Total Number of Patients	N	303	313		
Technical Success (Corelab Quali)$	n/N	243/303	257/313		
	% (95% CI)	80.2 (75.7; 84.7)	82.1 (77.9; 86.4)	-1.9 (-8.1; 4.3)	0.5443
Technical Success (Corelab QCA)$	n/N	226/301	232/312		
	% (95% CI)	75.1 (70.2; 80.0)	74.4 (69.5; 79.2)	0.7 (-6.2; 7.6)	0.8366
Technical Success of Main Branch (Corelab Quali)$	n/N	243/303	258/311		
	% (95% CI)	80.2 (75.7; 84.7)	83.0 (78.8; 87.1)	-2.8 (-8.9; 3.4)	0.3776

		Randomized treatment			
Secondary Procedural Endpoints	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Difference (95% CI)**	**P-value**
Technical Success of Main Branch (Corelab QCA)$	n/N	226/301	233/310		
	% (95% CI)	75.1 (70.2; 80.0)	75.2 (70.4; 80.0)	-0.1 (-6.9; 6.8)	0.9822
Technical Success of Side Branch#	n/N	303/303	310/313		
	% (95% CI)	100 (98.8; 100)	99.0 (97.2; 99.8)	1.0 (-0.3; 2.8)	0.2490
Procedure time [min]&	N	303	313		
	Median (Q1,Q3)	34.0 (25.0; 50.0)	45.0 (33.0; 62.0)	-10.0 (-13.0; -7.0)	<.0001
Fluoroscopic Time [min]&	N	303	313		
	Median (Q1,Q3)	10.0 (6.3; 15.8)	13.2 (9.0; 20.0)	-3.3 (-4.4; -2.3)	<.0001
Total Radiation Dose (Air Kerma) [mGy]&	N	293	303		
	Median (Q1,Q3)	453.4 (240.0; 872.0)	629.0 (332.0; 1025)	-113 (-180; -50.3)	0.0005
Total Contrast Volume [mL]&	N	303	313		
	Median (Q1,Q3)	130.0 (100.0; 170.0)	150.0 (120.0; 190.0)	-23.0 (-30.0; -15.0)	<.0001
Need to cross-over$	n/N	6/303	5/313		
	% (95% CI)	2.0 (0.4; 3.5)	1.6 (0.2; 3.0)	0.4 (-1.7; 2.5)	0.7199

n: number of patients with event, N: number of patients with data, CI: 2-sided confidence interval, SD: standard deviation.
$ Analysed using chi-square test.
# Analysed using Fisher exact test. Confidence intervals are exact confidence intervals.
@ Analysed using 2-sample t-test.
& Analysed using Wilcoxon rank-sum test. Differences (CI) are estimated by the Hodges-Lehmann estimator for the median difference.

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Secondary clinical endpoints are summarized in Table 18. No significant differences were observed between the SB and no-SB intervention groups with respect to Target Lesion Failure or any of its individual components (Cardiac Death, Target Vessel Myocardial Infarction, and Clinically Driven Target Lesion Revascularization). Similarly, there were no significant differences between groups regarding Main Branch or Side Branch TLR, nor with respect to stent thrombosis.

At the 12-month follow-up, the improvement in angina status was significantly higher in the no-SB intervention group (99%) compared with the SB intervention group (93.7%).

Further statistics describing Secondary Clinical Endpoints are provided in the statistical reports. (Appendix 9, Table 10).

Table 18. Secondary Clinical Endpoints ITT

		Randomized treatment			
Secondary Clinical Endpoints	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Difference (95% CI)**	**P-value**
Total Number of Patients	N	303	313		
Target Lesion Failure					
At 1 Month	n	11	14		
	% (95% CI)	3.6 (1.9; 6.2)	4.5 (2.6; 7.2)	-0.8 (-4.0; 2.3)	0.5967
At 1 Year	n	15	20		
	% (95% CI)	5.0 (2.9; 7.8)	6.4 (4.0; 9.5)	-1.4 (-5.1; 2.2)	0.4418
Cardiac Death					
At 1 Month	n	0	0		
	% (95% CI)	0.0	0.0	0.0	NC

Secondary Clinical Endpoints	Statistic	Randomized treatment			
		No side branch intervention	Side branch intervention	Difference (95% CI)	P-value
At 1 Year	n	1	0		
	% (95% CI)	0.3 (0.0; 1.8)	0.0	0.3	NC
Target vessel MI					
At 1 Month	n	10	14		
	% (95% CI)	3.3 (1.7; 5.8)	4.5 (2.6; 7.2)	-1.2 (-4.2; 1.9)	0.4521
At 1 Year	n	12	15		
	% (95% CI)	4.0 (2.2; 6.6)	4.8 (2.8; 7.6)	-0.8 (-4.1; 2.4)	0.6170
Clinically driven TLR					
At 1 Month	n	2	0		
	% (95% CI)	0.7 (0.1; 2.2)	0.0	0.7	NC
At 1 Year	n	4	6		
	% (95% CI)	1.3 (0.4; 3.2)	1.9 (0.8; 4.0)	-0.6 (-2.6; 1.4)	0.5557
Death					
At 1 Month	n	1	1		
	% (95% CI)	0.3 (0.0; 2.3)	0.3 (0.0; 2.3)	0.0 (-0.9; 0.9)	0.9816
At 1 Year	n	6	7		
	% (95% CI)	2.0 (0.9; 4.4)	2.3 (1.1; 4.8)	-0.3 (-2.6; 2.0)	0.7954
Main Branch TLR					
At 1 Month	n	2	0		

		Randomized treatment			
Secondary Clinical Endpoints	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Difference (95% CI)**	**P-value**
	% (95% CI)	0.7 (0.1; 2.2)	0.0	0.7	NC
At 1 Year	n	4	6		
	% (95% CI)	1.3 (0.4; 3.2)	1.9 (0.8; 4.0)	-0.6 (-2.6; 1.4)	0.5557
Side Branch TLR					
At 1 Month	n	1	0		
	% (95% CI)	0.3 (0.0; 1.7)	0.0	0.3	NC
At 1 Year	n	2	1		
	% (95% CI)	0.7 (0.1; 2.2)	0.3 (0.0; 1.7)	0.3 (-0.8; 1.5)	0.5473
Definite or Probable Stent Thrombosis					
At 1 Month	n	1	1		
	% (95% CI)	0.3 (0.0; 1.7)	0.3 (0.0; 1.7)	0.0 (-0.9; 0.9)	0.9816
At 1 Year	n	2	2		
	% (95% CI)	0.7 (0.1; 2.2)	0.6 (0.1; 2.2)	0.0 (-1.3; 1.3)	0.9712
Improvement in Angina Status					
At 1 Month	n/N	101/106	104/117		
	% (95% CI)	95.3 (91.2; 99.3)	88.9 (83.2; 94.6)	6.4 (-0.6; 13.4)	0.0800
At 12 Months	n/N	102/103	104/111		
	% (95% CI)	99.0 (97.1; 100)	93.7 (89.2; 98.2)	5.3 (0.4; 10.2)	0.0398

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6.4.9 Primary Endpoint by subgroup

Primary endpoint (PPMI) results by subgroups are presented in Table 19. The PPMI rate in the no-SB intervention group was significantly non-inferior across most of the tested subgroups: sex, age, and side branch diameter. In the Medina X,X,1 class subgroup, the PPMI rate was not non-inferior (even though the results between the groups were very close). Additionally, when the stenosis percentage in the side branch exceeded 50%, the PPMI rate in the no-SB intervention group was also not non-inferior. However, no significant p-value for interaction was observed in between subgroups, meaning that the outcomes for the primary endpoint was consistent in all the pre-specified subgroups.

Further statistics describing Primary Endpoint analysis by subgroup are provided in the statistical reports. (Appendix 9, Table 11).

Table 19. Primary Endpoint (PPMI) by subgroup_ITT set

		Randomized treatment			
Primary Endpoint	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Risk Difference (95% CI)**	**P-value**
Total Number of Patients	N	303	313		
Total Population	n/N	11/273	16/282		
	% (95% CI)	4.1% (1.7; 6.4)	5.7% (2.9; 8.4)	-1.6% (-∞; 1.4)	<.0001$
Sex (Interaction: p=0.8046)					
Female	n/N	2/ 65	5/ 60		
	% (95% CI)	3.2% (-1.1; 7.6)	8.0% (1.2; 14.9)	-4.8% (-∞; 2.0)	0.0085$

		Randomized treatment			
Primary Endpoint	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Risk Difference (95% CI)**	**P-value**
Male	n/N	9/208	11/222		
	% (95% CI)	4.3% (1.6; 7.1)	5.0% (2.1; 7.9)	-0.7% (-∞; 2.7)	0.0006$
Age (Interaction: p=0.6985)					
<70 years	n/N	4/146	9/151		
	% (95% CI)	2.9% (0.1; 5.7)	5.9% (2.2; 9.7)	-3.0% (-∞; 0.9)	0.0003$
>= 70 years	n/N	7/127	7/131		
	% (95% CI)	5.5% (1.5; 9.4)	5.4% (1.5; 9.3)	0.1% (-∞; 4.7)	0.0190$
Medina Class (Interaction: p=0.8919)					
1,1,0 & 1,0,0 & 0,1,0	n/N	5/189	9/182		
	% (95% CI)	2.9% (0.4; 5.3)	5.0% (1.8; 8.3)	-2.2% (-∞; 1.2)	<.0001$
1,1,1 & 1,0,1 & 0,1,1	n/N	6/ 84	7/100		
	% (95% CI)	6.9% (1.5; 12.2)	6.8% (1.9; 11.7)	0.1% (-∞; 6.2)	0.0719$
Angiographic Diameter of Side Branch by Corelab (Interaction: p=0.4749)					
<= 2.25 mm	n/N	4/105	3/122		
	% (95% CI)	3.9% (0.3; 7.6)	2.7% (-0.3; 5.8)	1.2% (-∞; 5.1)	0.0342$
> 2.25 mm	n/N	7/165	13/157		
	% (95% CI)	4.2% (1.2; 7.3)	7.9% (3.8; 12.1)	-3.7% (-∞; 0.7)	0.0002$

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		Randomized treatment			
Primary Endpoint	**Statistic**	**No side branch intervention**	**Side branch intervention**	**Risk Difference (95% CI)**	**P-value**
Side Branch Residual Stenosis by Corelab QCA (Interaction: p=0.2110)					
<= 50%	n/N	5/209	14/249		
	% (95% CI)	2.6% (0.4; 4.8)	5.7% (2.8; 8.6)	-3.1% (-∞; -0.0)	<.0001$
> 50%	n/N	6/ 62	1/ 31		
	% (95% CI)	9.1% (2.2; 16.0)	3.3% (-3.0; 9.5)	5.8% (-∞; 13.7)	0.7235$

n= number of events; N= number of patients in the analysis, $= one-sided non-inferiority test with a margin of 7.56%.
Missing data were accounted for by performing 100 multiple imputations.

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6.4.10 Safety evaluation

Summary of adverse events and adverse device effects

Details of adverse events and adverse device effects are listed in Annual Safety Report, final version dated November 23rd, 2023, available in Appendix 10.

7 Discussion

The KISS trial is the first randomized study that investigated the impact of no SB intervention in a common population of coronary patients treated for non-LM bifurcation coronary lesions. The main findings can be summarized as follows. Firstly, the intentional non-SB intervention strategy is non-inferior compared to a systematic SB intervention strategy and resulted in numerically lower rates of PPMI. Secondly, a non-SB intervention strategy is associated with simpler procedures as demonstrated by a significant reduction of procedure time, X-ray dose and volume contrast. Thirdly, in a contemporary cohort of bifurcation PCI, the rate of procedural complications remains very low and without significant differences between the two intervention strategies, while a higher rate of SB dissection is observed in case of systematic SB intervention. Likewise, the rate of major adverse clinical events and mortality are very low without difference according to PCI strategy.

Lastly, although it is a surrogate and subjective endpoint, a significant angina improvement at one year is observed in the group of patients without systematic SB intervention.

Periprocedural myocardial infarctions are a continuing topic of debate considering its best definition and clinical relevance. Although there are often asymptomatic, these periprocedural complications can delay hospital discharge. More importantly, several studies and metanalyses have reported that periprocedural myonecrosis, even of limited magnitude, is associated with late adverse cardiac events and all-cause mortality (9,10). Thus, even if the present study was not powered to evaluate long-term clinical events, the significant reduction of periprocedural myocardial injuries and infarctions associated to the conservative treatment of SB without systematic intervention appears relevant with clinical impact.

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Consequently, as supported by our results, the POT is the only recommended intervention to treat coronary bifurcation after stenting of the main vessel (6,8) and intervention on the SB appears only optional in case of flow impairment or acute occlusion of the SB. In these cases, the optimal revascularization strategy remains uncertain. The present study suggests a better result of only SB ballooning in comparison to kissing balloon inflation with a significant reduction of peri-procedural MI and injuries (significant p-value for non-inferiority and a trend for superiority). It is still important to note that it was a sub-group analysis and that the choice of SB intervention technique was left at discretion of practitioner without randomization. Although only hypothesis-generating, these data are consistent with previous observational study that underlined the safety and efficacy of SB ballooning with only 2% of one-year target lesion failure(24).

In more complex anatomical situations, a two-stent strategy for bifurcation revascularization should be considered. Several trials aimed to evaluate the benefit of its strategy in comparison to a provisional stenting with one-stent planed strategy (25–29). It appears that a two-stent strategy should be limited to complex bifurcation including especially a pathological SB with ostial lesion longer than 10mm, extensive calcifications or predicted difficulty for rewiring(4,6,30). A multivariate risk score recently published confirmed these criteria with an higher rate of combined all-cause death, MI and TVR in case of SB thrombus, SB moderate/severe calcium, and SB lesion length >10 mm(31). However, these situations are relatively rare in daily clinical practice (<10%)(4,6) and a provisional stenting approach remains the preferred default technique for most of bifurcation lesions in routine clinical practice.

The population included in the present study appears representative of daily clinical practice and consistent with previous studies with a predominance of LAD/diagonal bifurcation, more than 50% of patients with “true bifurcation”, a limited disease of the SB with a median SB lesion length lower than 5mm. Its findings may thus be particularly relevant to everyday practice, where most bifurcations are non-complex but the results cannot be extrapolated for the management of LM due to exclusion of these patients in this study. Due to larger diameters and lower subsequent risk of in-stent restenosis or plaque shifting but also the inacceptable compromise result for the “side branch”, PCI of the LM is commonly considered with specific management. However, the last studies that evaluated PCI strategy for LM provide similar conclusions than non-LM bifurcations with fewer major adverse cardiac events and long term TLR in case of provisional stenting strategy(32–34). Although no study has specifically addressed this issue, a KISS strategy for LM may be beneficial particularly due to lower risk of flow impairment in a large SB and the limited risk of SB rewiring in case of reintervention due to larger stent struts.

Several limitations of the KISS trial should be acknowledged. First, although the study was based on a randomized, multicentric and international design, it remained

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open-label, which may introduce operator biases in procedural decisions and follow-up assessments. Nevertheless, as the other previous studies on this setting, due to the evaluation of the intervention on SB, a double blinded study appears not really feasible. Moreover, all procedures were reviewed by a corelab with two independent physicians. Second, the inclusion criteria focused on bifurcation lesions with preserved SB flow, could potentially limit the generalizability of the findings to more complex or high-risk anatomies. Third, the definition of periprocedural MI, while aligned with ARC-2 criteria, relies partly on biomarker-based MI definition, which can be influenced by procedural variables and lacks standardization across centres. Fourth, the trial follow-up duration and the lower rate of observed MI than expected may limit its power to detect differences between the groups. Finally, the PCI was not guided by intracoronary imaging despite the demonstrated significant reduction of major adverse cardiac events in complex PCI(35,36,37). Nevertheless, the evidence of this clinical benefit, particularly for the treatment of bifurcation lesions and the subsequent international guidelines have been published after the designing of the trial and the patients' inclusion.

8 Overall conclusions

The management of the SB during bifurcation PCI remains challenging. A conservative strategy without systematic SB intervention in case of normal SB flow is associated with very rare procedural complications and is non inferior to a systematic SB intervention regarding periprocedural MI and injuries. These findings support a simplified "Keep Bifurcation Stenting Simple" approach in non-complex and non-LM bifurcation PCI. Further studies are warranted to support these data and evaluate the impact of this conservative strategy on long-term clinical events.

9 Abbreviations and definitions

9.1 Abbreviations

Abbreviation	Term
ACC/AHA	American College of Cardiology/American Heart Association
ARC	Academic Research Consortium
AT	As Treated
BMS	Bare Metal Stent
CABG	Coronary Artery Bypass Graft
CE	Conformité Européenne

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CEC	Clinical Events Committee
CERC	Cardiovascular European Research Center
CERIC	Center for European Research Initiatives in Cardiovascular Medicine
CRA	Clinical Research Associate
DAPT	Antiplatelet Therapy
DES	Drug Eluting Stent
eCRF	Electronic Case Report Form
EC	Ethics Committee
ECG	Electrocardiogram
EDC	Electronic Data Capture
FDA	Food and Drug Administration
GCP	Good Clinical Practices
hs-cTnI	high-sensitivity cardiac troponin I
hs-cTnT	high-sensitivity cardiac troponin T
ICF	Informed Consent Form
IFU	Instruction for Use
ITT	Intent-to treat
IVUS	Intravascular Ultrasound
KBT	Kissing Balloon Technique
KBI	Kissing Balloon Inflation
LAD	Left Anterior Descending artery
MB	Main Branch
MI	Myocardial Infarction
MV	Monitoring Visit
NOAC	Non–Vitamin K Oral Anticoagulant
OCT	Optical Coherence Tomography
OTW	Over-the-Wire
PCI	Percutaneous Coronary Intervention
POT	Proximal Optimization Technique
PPMI	Peri-Procedural Myocardial Infarction
QCA	Quantitative Coronary Analysis
RX	Rapide Exchange
SAP	Statistical Analysis Plan
SB	Side Branch
SDV	Source Data Verification

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TIMI	Thrombolysis In Myocardial Infarction
TLF	Target Lesion Failure
TLR	Target Lesion Revascularization
TVR	Target Vessel Revascularization
PP	Per Protocol
VKA	Vitamin K Antagonist

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9.2 Definitions

Term	Definition
ACUTE VESSEL OCCLUSION	The state of complete luminal obstruction with no antegrade blood flow
ANGINA STATUS (CCS (21))	**Grade 1:** "Ordinary physical activity does not cause ... angina," such as walking and climbing stairs. Angina with strenuous or rapid or prolonged exertion at work or recreation. **Grade 2:** "Slight limitation of ordinary activity." Walking or climbing stairs rapidly, walking uphill, walking or stair climbing after meals, or in cold, or in wind, or under emotional stress, or only during the few hours after awakening. Walking more than 2 blocks on the level and climbing more than one flight of ordinary stairs at a normal pace and in normal conditions. **Grade 3:** "Marked limitation of ordinary physical activity." Walking one to two blocks on the level and clim-bing one flight of stairs in normal conditions and at normal pace. **Grade 4:** "Inability to carry on any physical activity without discomfort - anginal syndrome may be present at rest."
BLEEDING (BARC (38))	**Life-threatening or Disabling Bleeding** • Fatal bleeding (Bleeding Academic Research Consortium [BARC] type 5) OR • Bleeding in a critical organ, such as intracranial, intraspinal, intraocular, or pericardial necessitating pericardiocentesis, or intramuscular with compartment syndrome (BARC type 3b and 3c) OR • Bleeding causing hypovolemic shock or severe hypotension requiring vasopressors or surgical intervention (BARC type 3b) OR • Overt source of bleeding with drop in hemoglobin of ≥5 g/dL or whole blood or packed red blood cells (RBC) transfusion ≥4 units (BARC type 3b)* **Major Bleeding (BARC type 3a)** • Overt bleeding either associated with a drop in the hemoglobin level of at least 3.0g/dL or requiring transfusion of 2 or 3 units of whole blood/RBC, or causing hospitalization or permanent injury, or requiring surgical intervention AND does not meet criteria of life-threatening

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Term	Definition
	or disabling bleeding **<u>Minor Bleeding (BARC type 2 or 3a, depending on the severity)</u>** • Any bleeding worthy of clinical mention (e.g., access site hematoma) that does not qualify as life-threatening, disabling, or major ** Given one unit of packed RBC typically will raise blood hemoglobin concentration by 1 g/dL, an estimated decrease in hemoglobin will be calculated.*
CARDIAC DECOMPENSATION	Inability of the heart to maintain adequate circulation.
CARDIOGENIC SHOCK	An insufficient forward cardiac output to maintain adequate perfusion of vital organs to meet ongoing demands for oxygenation and metabolism. Cardiogenic shock is due to either inadequate left ventricular pump function (such as in congestive heart failure) or inadequate left ventricular filling (such as in cardiac tamponade). Cardiogenic shock is defined as sustained hypotension (>30 minutes) with evidence of tissue hypoperfusion including oliguria (<30 mL/h), cool extremities, cyanosis, and altered mental status.
CEREBRAL INFARCTION	Evidence of brain cell death from imaging studies or pathological examination. If there are clinical symptoms, then it is a stroke; otherwise, it is an asymptomatic cerebral infarction.
CORONARY OBSTRUCTION	Angiographic or echocardiographic evidence of a new, partial or complete, obstruction of a coronary ostium, by the native leaflets, calcifications, or dissection, occurring during or after the procedure. The diagnosis of coronary obstruction can be determined by imaging studies (coronary angiography, intravascular ultrasound, multi-slice computed tomography angiography, or echocardiography), surgical exploration, or autopsy findings. Cardiac biomarker elevations and ECG changes indicating new ischemia provide corroborative evidence.
DEATH (ARC-2 (20))	**<u>Cardiovascular death</u>** Defined as death resulting from cardiovascular causes. The following categories may be collected: 1. Death caused by acute MI

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Term	Definition
	2. Death caused by sudden cardiac, including unwitnessed, death 3. Death resulting from heart failure 4. Death caused by stroke 5. Death caused by cardiovascular procedures 6. Death resulting from cardiovascular hemorrhage 7. Death resulting from other cardiovascular Cause **Noncardiovascular death:** Defined as any death that is not thought to be the result of a cardiovascular cause. The following categories may be collected: 1. Death resulting from malignancy 2. Death resulting from pulmonary causes 3. Death caused by infection (includes sepsis) 4. Death resulting from gastrointestinal causes 5. Death resulting from accident/trauma 6. Death caused by other noncardiovascular organ failure 7. Death resulting from other noncardiovascular cause **Undetermined** Defined as a death not attributable to any other category because of the absence of any relevant source documents. Such deaths will be classified as cardiovascular for end point determination.
MYOCARDIAL INFARCTION (MI) (ARC-2) (20)	**Criteria for acute myocardial infarction** The term acute myocardial infarction (MI) should be used when there is evidence of myocardial necrosis in a clinical setting consistent with acute myocardial ischemia. Under these conditions any one of the following criteria meets the diagnosis for MI: ● Detection of a rise and/or fall of cardiac biomarker values [preferably cardiac troponin (cTn)] with at least one value above the 99th percentile upper reference limit (URL) and with at least one of the following: - Symptoms of ischemia. - New or presumed new significant ST-segment–T wave (ST–T) changes or new left bundle branch block (LBBB). - Development of pathological Q waves in the ECG. - Imaging evidence of new loss of viable myocardium or

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Term	Definition
	new regional wall motion abnormality. - Identification of an intracoronary thrombus by angiography or autopsy. ● Cardiac death with symptoms suggestive of myocardial ischemia and presumed new ischemic ECG changes or new LBBB, but death occurred before cardiac biomarkers were obtained, or before cardiac biomarker values would be increased. ● Percutaneous coronary intervention (PCI) related MI is arbitrarily defined by elevation of cTn values (>5 x 99th percentile URL) in patients with normal baseline values (≤99th percentile URL) or a rise of cTn values >20% if the baseline values are elevated and are stable or falling. In addition, either (i) symptoms suggestive of myocardial ischemia or (ii) new ischemic ECG changes or (iii) angiographic findings consistent with a procedural complication or (iv) imaging demonstration of new loss of viable myocardium or new regional wall motion abnormality are required. ● Stent thrombosis associated with MI when detected by coronary angiography or autopsy in the setting of myocardial ischemia and with a rise and/or fall of cardiac biomarker values with at least one value above the 99th percentile URL. ● Coronary artery bypass grafting (CABG) related MI is arbitrarily defined by elevation of cardiac biomarker values (>10 x 99th percentile URL) in patients with normal baseline cTn values (≤99th percentile URL). In addition, either (i) new pathological Q waves or new LBBB, or (ii) angiographic documented new graft or new native coronary artery occlusion, or (iii) imaging evidence of new loss of viable myocardium or new regional wall motion abnormality. **Criteria for prior myocardial infarction** Any one of the following criteria meets the diagnosis for prior MI: ● Pathological Q waves with or without symptoms in the absence of non-ischemic causes. ● Imaging evidence of a region of loss of viable myocardium that is thinned and fails to contract, in the absence of a non-ischemic cause. ● Pathological findings of a prior MI.

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Term	Definition
PERI-PROCEDURAL MI (ARC-2) (20)	Absolute rise in cardiac troponin (from baseline) ≥35 times upper reference limit + 1 (or more) of the following criteria: - New significant* Q waves or equivalent - Flow-limiting angiographic complications - New “substantial” loss of myocardium on imaging **Significant periprocedural myocardial injury:** Absolute rise in cardiac troponin (from baseline) ≥70 times upper reference limit *Definitions applicable in patients with normal (or elevated and stable or falling) baseline biomarkers. *Q-wave criteria requires the development of new Q waves ≥40 ms in duration and ≥1 mm deep in voltage in ≥2 contiguous leads*
PROCEDURE-RELATED EVENTS	Events occurring during or as a direct result of the index procedure.
SOURCE DATA (per ISO 14155:2011)	All information in original records of clinical findings, observations, or other activities in a clinical investigation, necessary for the reconstruction and evaluation of the clinical investigation.
SOURCE DOCUMENT (per ISO 14155:2011)	Printed, optical or electronic document containing source data. Examples: Hospital records, laboratory notes, device accountability records, photographic negatives, radiographs, records kept at the investigation center, at the laboratories and at the medico-technical departments involved in the clinical investigation.
STENT THROMBOSIS (ARC-2) (20)	**Definite stent thrombosis:** Angiographic confirmation of stent thrombosis* • The presence of a thrombus† that originates in the stent/scaffold or in the segment 5 mm proximal or distal to the stent/scaffold or in a side branch originating from the stented segment and the presence of at least 1 of the following criteria: - Acute onset of ischemic symptoms at rest, - New electrocardiographic changes suggestive of acute ischemia, - Typical rise and fall in cardiac biomarkers (refer to definition of spontaneous myocardial infarction),

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Term	Definition
	OR • Pathological confirmation of stent thrombosis - Evidence of recent thrombus within the stent determined at autopsy - Examination of tissue retrieved following thrombectomy (visual/histology) **Probable stent thrombosis:** Regardless of the time after the index procedure, any myocardial infarction that is related to documented acute ischemia in the territory of the implanted stent/scaffold without angiographic confirmation of stent thrombosis and in the absence of any other obvious cause.‡ **Silent stent occlusion:** The incidental angiographic documentation of stent occlusion in the absence of clinical signs or symptoms is not considered stent thrombosis. **Timing of Stent thrombosis** (duration after stent implantation): - Acute: 0§–24 hours - Subacute: >24 h–30 days - Late: >30 d–1 year - Very late: >1 year *Early stent thrombosis is 0 to 30 days (acute plus subacute stent thrombosis).* **Definite stent/scaffold thrombosis is considered to have occurred by either angiographic or pathological confirmation.* *†Occlusive thrombus: Thrombolysis in Myocardial Infarction grade 0 or 1 flow within or proximal to a stent/scaffold segment. Nonocclusive thrombus: intracoronary thrombus is defined as a (spherical, ovoid, or irregular) noncalcified filling defect or lucency surrounded by contrast material (on 3 sides or within a coronary stenosis) seen in multiple projections, persistence of contrast material within the lumen, or visible embolization of intraluminal material downstream.* *‡When the stented/scaffolded segment is in the left circumflex coronary artery or in the presence of preexisting electrocardiographic abnormalities (eg, left bundle branch block, paced rhythms), definitive evidence of localization may be absent and Clinical Events Committee adjudication is based on review of all available evidence).* *§Defined as the moment the patient is undraped and taken off the catheterization table.*
STROKE	Sudden onset of vertigo, numbness, dysphasia, weakness, visual field defects, dysarthria or other focal neurological deficits due to vascular lesions of the brain such as hemorrhage, embolism, thrombosis, or rupturing aneurysm, that persists >24 hours.
TARGET LESION	Treated lesion starting 5 mm proximal of the stented lesion

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Term	Definition
	and to end 5 mm distal of the stented lesion.
TARGET LESION FAILURE	Composite of: Cardiac Death, Target vessel Myocardial Infarction, Clinically driven Target lesion revascularization
TARGET LESION RE-VASCULARIZATION	Any repeat percutaneous intervention of the target lesion or bypass surgery of the target vessel. Clinically-driven revascularizations are those in which the patient has a positive functional study, ischemic ECG changes at rest in a distribution consistent with the target vessel, or ischemic symptoms. Revascularization of a target lesion with an in-lesion diameter stenosis ≥70% (by QCA) in the absence of the above-mentioned ischemic signs or symptoms is also considered clinically-driven. In the absence of QCA data for relevant follow-up angiograms, the clinical need for revascularization is adjudicated using the presence or absence of ischemic signs and symptoms. Non-clinically driven repeat target lesion revascularizations are those in which the patient undergoes a non-emergent revascularization for a diameter stenosis <50% (by QCA). Non-emergent repeat target lesion revascularization for a diameter stenosis <70% (by QCA) in patients without either a positive functional study or angina are also considered non-clinically driven defined as any repeat revascularization of the target site whether by PCI or bypass surgery.
TARGET VESSEL	Index coronary artery which was in physical contact with any component (guiding catheter, guide wire, balloon catheter, etc.) of the angioplasty hardware during the initial procedure.
TARGET VESSEL RE-VASCULARIZATION	Any target vessel revascularization, death, or MI attributed to the target vessel.

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Signature Page

I have read this report and confirm that to the best of my knowledge it accurately describes the conduct and results of the study

SPONSOR'S REPRESENTATIVE:

Name: Signé par : Stephane Cook

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Signature, date

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CRO'S REPRESENTATIVE:

Name: Peter Smits

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Peter Smits

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Signature, date

11-Feb-26

COORDINATING INVESTIGATOR SIGNATURE:

Name: Bernard CHEVALIER

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11 Appendices

- Appendix #1: Clinical Investigation Plan
- Appendix #2: Corelab Plan
- Appendix #3: Master Copy of the Inform Consent Form
- Appendix #4: Monitoring Plan
- Appendix #5: Quality Plan (Appendix 3 of the Quality Plan)
- Appendix #6: Data Management Plan
- Appendix #7: Statistical Analysis Plan
- Appendix #8: CEC Charter
- Appendix #9: Statistical Analysis Report
- Appendix #10: Annual Safety Report Final version